Detection of Externally-Influenced Routing via Inverse Markov-Chain Methods

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Imperial College London
MEng Design Engineering, Master's Project

June 2026

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Abstract

Satellite-navigation and live-traffic services increasingly shape urban route choice by reallocating guided drivers around congestion. This weakens the revealed-preference assumption behind demand estimation, route-choice modelling and transport policy: observed traffic may reflect a mixture of intrinsic driver preferences and external routing recommendations. This dissertation develops a partial-observation routing-influence filter for that setting. Given a road graph, sparse observation stations and two traffic regimes, the filter fits one inverse Markov-chain model per regime and scores each trajectory with a log-likelihood ratio. The route-choice estimator is the partial-observation inverse Markov-chain method of Morimura, Osogami and Idé [1]; the contribution here is the two-regime filter, its trajectory-level scoring rule and its diagnostic tests. A compact primer on the mathematical objects and evaluation metrics is provided in section F.

The filter is evaluated in three tiers. Tier 1 reproduces Morimura et al.'s synthetic recovery benchmark, validating the estimator implementation. Tier 2 uses controlled SUMO simulation with labelled sat-navigation rerouting. On long trajectories the fitted detector reaches an area under the receiver-operating-characteristic curve (AUC) of 0.708, compared with a full-observation empirical 1-step reference of 0.801. Feature regularisation improves per-row chain recovery across five SUMO seeds, although its aggregate AUC lift is not robust. A regularised empirical 2-step test lifts AUC by about +0.09 to +0.10, indicating higher-order route memory not captured by the current 1-step model. Tier 3 applies the same filter to 10.5 million reconstructed real automatic-vehicle-identification trips from Xuancheng (China), using rush-hour versus off-peak and Labour-Day versus normal travel as unlabelled behavioural proxies while controlling origin-destination mix. On rush-vs-off-peak, the fitted 1-step detector performs no better than random ranking (AUC near 0.5) across all tested zone granularities, even where the full-observation reference retains a weak signal. On the stronger Labour-Day split, full observation still separates

Labour-Day from normal trips after matching origin-destination cells (AUC = 0.567), but multistart validation-selected fitting falls back to random-ranking performance (AUC = 0.502), and an empirical 2-step audit does not improve the reference. The resulting contribution is a validated filter for controlled labelled settings and a clear account of its current limits on unlabelled real-world traffic.

Use of generative AI

Generative AI was used in this project as a coding and writing-support aid. LLM-based assistants, including Claude, Codex and Gemini, supported implementation, test scaffolding, mathematical derivation checks, experimental argument checks, contextual investigation, and report editing. The author reviewed all accepted code and text, and retained responsibility for the research design, interpretation, repository contents, and final submission.

Acknowledgement

I sincerely thank my supervisor, Professor Robert Shorten, for his guidance, kindness and support throughout the project.

Contents

1 Notation

A consolidated reference for the mathematical objects used throughout this report.

Table 1: Key symbols.

Symbol	Meaning
$\mathcal{X}$, n	Full state space (directed road edges); $n = \lvert\mathcal{X}\rvert$.
$X_o \subset \mathcal{X}$	Subset of observed states at which counts are available.
$\mathrm{adj}(x)$	Set of legal successors of state x (out-neighbours in the directed graph).
$p_T(x' \mid x)$	One-step transition probability from x to x'; row x of the kernel P_T.
$p_I(x)$	Restart distribution on $\mathcal{X}$; after a restart, the next state is drawn from this row vector.
$\beta \in [0, 1]$	Continuation probability per step; together with p_I defines the restarted stationary mixture $P = \beta P_T + (1 - \beta)\,\mathbf{1}p_I^\top$.
π	Stationary distribution of P; obtained from $\pi^\top P = \pi^\top$, $\mathbf{1}^\top \pi = 1$.
$f(x)$	Empirical visit count (or normalised frequency) at x. Observed only for $x \in X_o$.
$g(x, x')$	Hitting rate from x to x', a discounted first-passage probability on $X_o \times X_o$; empirical in SUMO/Xuancheng and analytically generated only in the Tier 1 synthetic benchmark.
$\phi_T(x') \in \mathbb{R}^{d_T}$	Destination-state feature vector (section 2.4).
$\psi(x, x') \in \mathbb{R}^{d_\psi}$	Directed-edge feature vector.
$\nu \in \mathbb{R}^n$	Per-state base parameters of p_I.
$\omega^{\mathrm{loc}} \in \mathbb{R}^E$	Per-edge local weights of p_T, with $E = \sum_x \lvert\mathrm{adj}(x)\rvert$ the number of directed edges.
$\omega^{\mathrm{glo}_1} \in \mathbb{R}^{d_T}$	Shared weights on destination features ϕ_T (paper Eq. 17).
$\omega^{\mathrm{glo}_2} \in \mathbb{R}^{d_\psi}$	Shared weights on edge features ψ (paper Eq. 17).
θ	Full parameter vector $[\nu; \omega^{\mathrm{loc}}; \omega^{\mathrm{glo}_1}; \omega^{\mathrm{glo}_2}] \in \mathbb{R}^d$, $d = n + E + d_T + d_\psi$.
λ	Ridge/Tikhonov regularisation strength: the weight on the L_2 penalty $\lVert\theta\rVert^2$, used to discourage very large fitted parameters.
$\gamma \in [0, 1]$	Mixing weight between the stationary-match loss L_d and the hitting-rate loss L_h.
$\Lambda(\tau)$	Log-likelihood-ratio score of a test trajectory τ under two fitted chains (section 2.3).
AUC	Area under the ROC curve of the LR score against labelled regime test trajectories.
$\mathrm{RMAE}(\hat{\pi}, \pi)$	Relative mean absolute error of recovered stationary mass at unobserved states.

Throughout, "regime" refers to one of two distinguishable driver populations whose chains are fitted and compared. In controlled simulation this is intrinsic vs. sat-navigation-influenced; in real-world data it is a behavioural proxy (e.g. rush-hour vs. off-peak, or Labour Day vs. normal days). The contrast $\Delta_{\mathrm{fit}}(x, \cdot) = p_T^{(B)}(x, \cdot) - p_T^{(A)}(x, \cdot)$ between two fitted chains is the central quantity assessed against the empirical ground-truth contrast $\Delta_{\mathrm{emp}}(x, \cdot)$.

2 Introduction

2.1 Context and research gap

Satellite-navigation and live-traffic services have become part of the route-choice system rather than passive map displays. Travel information changes route-choice behaviour in congested networks [2, 3], and routing-app shortest-path guidance can redistribute traffic in ways that affect wider network performance, equilibrium costs and residential cut-through traffic [4–6]. This creates a measurement problem. Traffic counts and trajectories no longer necessarily reveal only the preferences of unaided drivers; they may also encode the recommendations of external routing systems reacting to congestion and to other guided vehicles. This is a closed-loop data problem: route recommendations alter the driver behaviour from which future route-choice models may then be estimated [7]. It motivates looking for changes in local vehicular movement patterns, in the spirit of Shorten and co-authors' junction-level pattern-identification work [8].

This matters because route-choice inference, demand estimation, pricing and infrastructure appraisal rely on revealed-preference data. If a substantial fraction of observed trips is externally guided, downstream models may mistake recommender-induced behaviour for intrinsic driver preference. The problem addressed here is therefore detection: given only sparse observation of a road network, the aim is to distinguish trajectories that look intrinsic from trajectories that look externally influenced. The practical end goal is a filter: flag or remove the externally influenced trajectories so that a later analyst can pass a cleaned, intrinsic-behaviour dataset to downstream route-choice, demand-estimation or policy-appraisal tasks.

Partial observation is the realistic version of this problem, not an artificial handicap. Many transport datasets see only selected roadside counts, AVI detections, or incomplete trip fragments rather than a full route history for every vehicle. Fixed-location volume sensors such as loops and cameras are sparse, and recent route-choice work therefore estimates behaviour from sparse flow measurements and speed data [9]. A filter that only works once every transition and app-use label is known would be useful as a benchmark, but not as an operational screening method.

There is also an industry motivation for making this distinction at the trajectory level. Insurers, fleet operators and transport analytics providers increasingly need to understand whether an observed driving event reflects a driver's own route choice or the influence of an external routing system. In the near term, that could support cleaner risk and exposure analysis for usage-based insurance or commercial fleet audits: for example, separating a driver who repeatedly chooses a high-risk route from a driver who was diverted there by live guidance. Longer term, the same distinction becomes part of a broader question in driving analytics: when a journey, near miss or disruption is shaped by AI-mediated navigation, the evidence should not be interpreted as if it came only from human preference.

2.2 Related work and novelty

Route choice is often represented as a sequence of link-level decisions. This sequential view sits within the wider dynamic discrete-choice tradition exemplified by Rust's Markov decision-process model [10], while transport route-choice models usually specialise the state to a road link and the decision to the next link. Multinomial logit and recursive-logit models parameterise the probability of taking the next road edge from the current edge [11–14]. In a one-step Markov-chain (MC) form, with road edges as states, the transition probability is

$$p_T(x' \mid x) = \frac{\exp(s(x, x'))}{\sum_{x'' \in \mathrm{adj}(x)} \exp(s(x, x''))}, \tag{1}$$

where $\mathrm{adj}(x)$ is the set of legal successors of edge x. The score $s(x, x')$ is the systematic utility of moving from x to x'. In the implementation used here it is a linear score made from a local edge parameter and optional shared network features,

$$s(x, x') = \omega^{\mathrm{loc}}_{(x,x')} + \phi_T(x')^\top \omega^{\mathrm{glo}_1} + \psi(x, x')^\top \omega^{\mathrm{glo}_2},$$

where $\phi_T(x')$ describes the destination edge and $\psi(x, x')$ describes the directed turn or road transition. The softmax converts these scores into probabilities over the legal outgoing edges (see sections F.1 and F.4 for the Markov-chain and softmax definitions used here). Markov-chain road-network dynamics have also been used by Shorten and co-authors as a compact representation for network-level regulation and control [15].

Existing work touches the filtering problem from several sides, but not the exact setting studied here. Traveller-information and routing-game papers model how information changes route choice, adoption or network performance, often treating guided and unguided users as known populations [2–5]. Trajectory-outlier methods detect unusual paths or sub-trajectories [16], and Gu et al. monitor junction-level turning pattern changes relative to a baseline [8]. Partial-observation route-choice and inverse-preference papers estimate route-choice models from incomplete trips, sparse sensor observations or inverse preference learning [17–19]. These are important adjacent problems, but they usually estimate one behavioural model, detect population-level anomalies, or assume richer trajectory observation than sparse station counts. The narrower gap here is a two-regime, partial-observation filter: fit one route-choice chain per regime from sparse observations, score individual trajectories under the two fitted chains, and audit whether the resulting detector reflects transition-level chain recovery rather than origin-destination (OD) mix or other aggregate confounds.

Morimura, Osogami and Idé introduced an inverse Markov-chain method for recovering a softmax-parametric chain from partial observations [1]. Their method fits a chain using counts at a subset of observed states X_o and, optionally, hitting rates between observed states. This dissertation keeps Morimura et al.'s estimator as the inverse-chain fitting engine and adds the two-regime filter around it: one chain is fitted per regime, each trajectory is scored by a log-likelihood ratio, and the experimental tiers report diagnostics that separate transition-level chain recovery from OD mix, optimiser basins and model-order error.

2.3 Filter specification and assumptions

The filter has two jobs. First, it learns one route-choice model for each regime from sparse station observations. Second, it scores each held-out trajectory by asking which fitted model makes that trajectory more likely. The input is a directed road graph $\mathcal{X}$, an observed subset $X_o \subseteq \mathcal{X}$, regime-labelled station counts f_A, f_B and held-out trajectories.

What is fitted. For each regime the filter fits a softmax-parametric MC ($\hat{P}_A$ and $\hat{P}_B$) by minimising Morimura et al.'s regularised log-ratio loss [1]. The count term asks the model to put the right relative stationary mass on observed stations. The optional hitting-rate term asks it to reproduce how often one observed station is reached after another. In the notation used here, the objective is

$$\begin{aligned} L(\theta) &= \gamma L_d(\theta) + (1-\gamma) L_h(\theta) + \lambda R(\theta), \\ L_d(\theta) &= \frac{1}{2} \sum_{i,j \in X_o} \left(\log \frac{\pi_i(\theta)}{\pi_j(\theta)} - \log \frac{f_i}{f_j} \right)^2, \\ L_h(\theta) &= \frac{1}{2} \sum_{i,j \in X_o} \left(\log H_{ij}(\theta) - \log g_{ij} \right)^2, \qquad R(\theta) = \frac{1}{2} \|\theta\|_2^2. \end{aligned} \tag{2}$$

The parameter vector is $\theta = [\nu; \omega^{\text{loc}}; \omega^{\text{glo}_1}; \omega^{\text{glo}_2}] \in \mathbb{R}^d$, with $d = n + E + d_T + d_\psi$. Here ν parametrises the restart distribution p_I, ω^{loc} gives one local weight per directed road transition, and the two ω^{glo} blocks are shared destination-edge and transition-feature weights. The term L_d matches observed station-count ratios through the model stationary distribution $\pi(\theta)$, L_h matches observed hitting rates g_{ij} through the model hitting rates $H_{ij}(\theta)$, γ controls the count-versus-hitting-rate balance, and λ is ridge regularisation, meaning an L_2 penalty that discourages very large parameter values and stabilises the under-constrained fit.

The full-network SUMO and Xuancheng fits use the f-only setting ($\gamma = 1$). Here "f-only" means fitting only to observed station counts, while "$f + g$" means also fitting the between-station hitting rates. The latter is not used in the full-network detector because empirical g is sparse and floor-dominated at city scale. In this context, "floor-dominated" means that most observed station pairs are almost never connected in the sample, so many hitting-rate estimates collapse to the same small minimum value (section 4.3.5). All parametric fits use L-BFGS-B (section F.11), because the objective is smooth, analytic gradients are available, and the parameter dimension is large enough that a limited-memory quasi-Newton method is a practical compromise between first-order gradient descent and full second-order Newton optimisation.

How trajectories are scored. The fitted chains $\hat{P}_A$ and $\hat{P}_B$ then define a log-likelihood-ratio (LLR) score

$$\Lambda(\tau) = \sum_t \left[\log \hat{p}_B(x_{t+1} \mid x_t) - \log \hat{p}_A(x_{t+1} \mid x_t) \right],$$

where positive values mean "more like regime B" and negative values mean "more like regime A". The log-likelihood ratio is used because it directly compares the two fitted generative models. Under correctly specified regime models, thresholding the likelihood ratio is the optimal binary test in the Neyman–Pearson sense [20], and in the Markov setting the score decomposes into per-transition evidence.

This choice also keeps the two experimental settings comparable. Posterior probabilities would require regime priors and calibration; single-chain likelihoods would not compare the two hypotheses directly; transition-distance scores would discard the full path probability; and discriminative classifiers would require labelled training trajectories that the real Xuancheng tier does not have. The LLR threshold can be adjusted for a desired false-positive/false-negative tradeoff without refitting either chain. Validation reports the area under the receiver-operating-characteristic curve (AUC) over possible decision thresholds (sections F.5 and F.6).

What assumptions this carries. The Morimura estimator carries assumptions that are used later in the report. First, it assumes a one-step MC: the current edge alone determines the next-edge distribution, while the full previous path sits outside the state. Second, its count term is a stationary count model. Up to an unknown exposure scale $c > 0$, the model treats observed counts as

$$f_i \approx c\,\pi_i(\theta), \qquad i \in X_o,$$

and Morimura's recovered count prediction has the form $\hat{f}(x) = \hat{c}\hat{\pi}(x)$; the scale calibration used in this dissertation is given in eq. (4). Its optional hitting-rate term treats g_{ij} as a β-discounted first-passage probability from observed station i to observed station j in the same stationary chain. This is an approximation for the four-hour SUMO rush window and for the Xuancheng time-of-day splits: their observed counts are time-integrals of changing demand, congestion, queue spillback and route composition rather than draws from a single invariant distribution. The stationary surrogate is retained because it is identifiable from sparse station counts and directly comparable to Morimura et al.; a fully time-dependent route-choice model would add assumptions that the available data cannot reliably support. The fitted chain should therefore be read as the best stationary, softmax-parametric projection of a non-stationary window under the Morimura log-ratio loss, with no claim that it is literal behavioural ground truth. Third, the softmax form restricts recoverable behaviour to the chosen feature and parameter family. The empirical full-observation transition matrix used for comparison is also a finite-data reference rather than an oracle.

2.4 Experimental structure

The report follows the experimental tiers directly. Tier 1 validates the inverse-MC estimator against Morimura et al.'s synthetic benchmark. Tier 2 uses Simulation of Urban MObility (SUMO) [21] to test the full filter under labelled sat-navigation rerouting. This controlled tier also carries the feature-regularised comparison, the multistart basin check and the empirical 2-step memory audit. Tier 3 applies the same filter to automatic vehicle identification (AVI) data from Xuancheng, China, where rush-hour versus off-peak and Labour-Day versus normal travel are used as unlabelled behavioural proxies for external routing influence. The Xuancheng tier is where OD-rebalancing is most important: real-world regime pools are matched by $(\text{origin zone}, \text{destination zone})$ before comparing their routing scores.

Algorithm 1 Partial-observation routing-influence filter used in the three tiers

Require: Road graph $\mathcal{X}$; observed stations X_o; two regime count vectors f_A, f_B; held-out trajectories $\{\tau_m\}$; optional features and hitting-rate estimates.

Ensure: Trajectory scores $\Lambda(\tau_m)$, AUC, and diagnostic checks.

1: Fit $\hat{P}_A$ from f_A on X_o by minimising eq. (2).

2: Fit $\hat{P}_B$ from f_B on X_o using the same model class and hyperparameter protocol.

3: **for** each held-out trajectory $\tau = (x_0, \ldots, x_T)$ **do**

4: Accumulate $\Lambda(\tau) = \sum_{t=0}^{T-1}[\log \hat{p}_B(x_{t+1} \mid x_t) - \log \hat{p}_A(x_{t+1} \mid x_t)]$.

5: **end for**

6: Rank trajectories by $\Lambda(\tau)$ and report AUC; choose a deployment threshold only if a specific false-positive/false-negative tradeoff is required.

7: Audit the score using the relevant tier checks: transition-contrast correlation, per-row cosine, OD rebalancing, multistart behaviour, and empirical 2-step memory.

Each tier reports scalar AUC together with the relevant trust audit: transition-contrast correlation, per-state cosine similarity, OD-matching sensitivity, multistart behaviour or the empirical 2-step reference, depending on the failure mode under test. Test trajectories are withheld from the empirical transition matrix and from the empirical-reference LLR scorer. The inverse-MC fit uses aggregate station counts for the whole regime window; those counts are not recomputed after removing held-out vehicles. The reported AUCs are therefore held out at the trajectory-scoring level rather than as a fully out-of-sample sensor-count experiment. The Discussion compares what generalises across tiers, the Conclusion states the supported claims, and the final section covers project management and reflection.

3 Tier 1: Validating the inverse-MC estimator inside the filter

Before testing sat-navigation detection, Tier 1 serves as an implementation check. The question is deliberately narrow: if the true Markov chain is known synthetically and only a subset of its states is observed, does this implementation recover the stationary distribution at the unobserved states about as well as Morimura et al.'s published method?

The reproduction follows Morimura et al.'s synthetic benchmark on a random $n = 100$ softmax-parametric chain and matches the published proposed-method RMAE curve to within ~ 0.05 for $|X_o| \geq 10$. RMAE is the relative mean absolute error of the recovered stationary mass on unobserved states; lower is better. The only material deviation is the $|X_o| = 5$ setting, where the 75/25 validation split leaves roughly one validation state and makes λ selection degenerate.

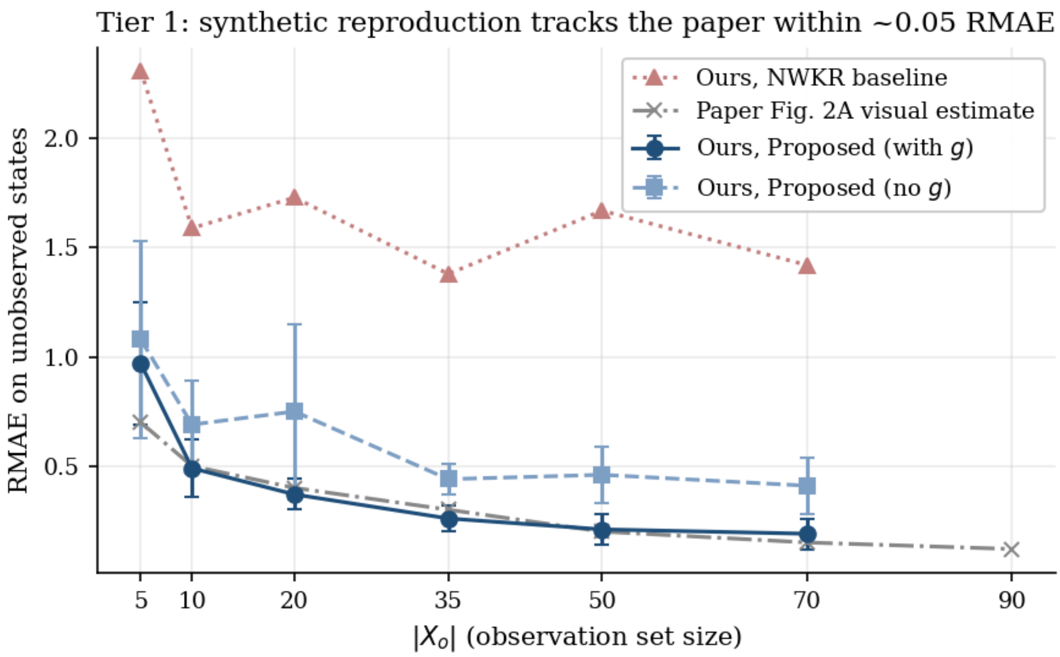

Figure 1: Tier 1 reproduction of Morimura et al.'s synthetic recovery benchmark. The vertical axis is RMAE on unobserved states, so lower is better. The reproduced proposed-method curve with hitting rates ("with g") follows the visually estimated paper curve closely once $|X_o| \geq 10$, which is the implementation check needed before using the inverse-MC estimator inside the two-regime filter.

The synthetic reproduction is therefore sufficient as an implementation gate for the downstream tiers: it verifies the analytic gradients, the partial-observation loss, and the estimator's stationary-distribution recovery under the benchmark's own assumptions. It does not by itself validate the later SUMO and Xuancheng detection settings. Full protocol details, the RMAE table, and the mathematical justification for the scale-free prediction convention are given in Section C in the Appendix.

4 Tier 2: Controlled validation of the filter in SUMO

4.1 Motivation

This tier is the controlled, end-to-end validation of the partial-observation routing-influence filter in Simulation of Urban MObility (SUMO) [21], the only setting here that supplies ground-truth regime labels: an identical population of vehicles is simulated once with sat-navigation rerouting turned off and once with it turned on. Every component of the filter is exercised at city-block scale ($n = 2{,}405$ edges): the inverse-MC estimator is fitted per regime, the log-likelihood-ratio scorer is applied to held-out trajectories, the feature-regularised filter variant is tested against a no-features baseline, and the filter's diagnostic suite (per-row Pearson/cosine, empirical 2-step memory audit and the appendix multistart check) is run on the resulting fits. A trajectory-length sweep traces how the scoring window T shifts both the empirical reference and the fitted detector together. The heavier multistart and rush/off-peak proxy audits are summarised in the main text and reported in Sections D and E in the Appendix. The section closes with the documented difficulty of using the hitting-rate $(f + g)$ extension at this scale, a design-relevant failure mode for the filter's optional g input.

A note on terminology used throughout this section. The no-rerouting regime is called "static unguided routing": SUMO routes those vehicles by static shortest-path Dijkstra with no live rerouting. Accordingly, "intrinsic" means the simulator's static-route baseline rather than a full behavioural model of human preference. The term "empirical 1-step reference" (rather than an information-theoretic ceiling) means the AUC obtained from the full-observation transition matrix after Laplace smoothing. Here, Laplace smoothing means adding a small pseudo-count to each legal transition before normalising, so legal but unseen transitions do not receive zero probability. The reference is therefore the best a 1-step detector can do given this finite data and smoothing, and finite-sample and train/test variation can move it (and occasionally produce apparent anomalies, section 5.4.1).

4.2 Method

4.2.1 Scenario construction

A rectangular Ingolstadt, Germany study area ($\sim 3.7\,\text{km} \times 3.3\,\text{km}$), derived from OpenStreetMap (OSM) via `netconvert`, yields $n = 2{,}405$ directed edges after filtering to the largest strongly-connected passenger-allowed component, with $E = 5{,}987$ directed transitions and mean out-degree ~ 2.5. A demand of $\sim 28{,}800$ vehicles is generated by `randomTrips.py` over a four-hour horizon with `-p 0.5`. In SUMO's `randomTrips.py`, `-p` is the average departure period in seconds, so `-p 0.5` attempts roughly one new trip every 0.5 s. A random seed is the integer that initialises this pseudo-random trip draw; changing it creates a different but comparable set of origins, destinations and insertion times. Each vehicle is replayed twice under the same insertion times and origin/destination pairs:

- **Intrinsic regime:** `-device.rerouting.probability 0`. Vehicles route via static shortest-path Dijkstra and never react to congestion.
- **Sat-navigation regime:** `-device.rerouting.probability 1`. Every vehicle re-evaluates its route every 30 s on the basis of observed live edge travel times.

Operating at `-p 0.5` was selected from preliminary demand sweeps: at higher demand both regimes saturate and the sat-navigation contrast collapses; at lower demand there is insufficient congestion to drive any rerouting. At `-p 0.5` the satnav regime delivers $\sim 36\%$ more completed trips than intrinsic with matched insertion (28,098 vs. 20,700 finishing). This completion imbalance is a limitation of the SUMO operating point. The validation scorer uses balanced test sets ($N = 300$ per regime), so the AUC cannot be explained by having more sat-navigation than intrinsic test vehicles. The count-fitting loss is based on station-count ratios, which removes dependence on the absolute total number of completed vehicles in each regime. Nevertheless, the fitted counts and the pool of long completed trajectories are conditional on different completion/survival processes in the two regimes. A real risk is therefore that the inverse-MC fit learns where trips survive to completion under sat-navigation rather than only learning routing preferences conditional on comparable trips. Tier 2 is therefore treated as a controlled operational validation of the filter under this biased operating point. It does not support a completion-balanced causal estimate of the effect of sat-navigation rerouting. A stricter sensitivity study would rerun the simulation or rebuild f under matched completed-trip populations; that extension is left outside the present experimental budget.

For each regime, the experiment exposes vehicle counts `edgedata.*.xml` (the empirical f) and complete per-vehicle routes `vehroutes.*.xml` (used as the test set and to construct the empirical chain for the contrast diagnostic). The inverter sees only $|X_o| = 601$ randomly sampled edges, 25% of the network, consistent with the operating point of the Tier 1 reproduction.

4.3 Results

4.3.1 Single-seed result: features and the inverse fit

The comparison uses two feature settings. `-features none` is the local-only model, with one parameter per directed transition ($d = n + E = 8{,}392$). `-features real` adds shared road-network features (road class, speed, lane count and turn geometry), giving $d = 8{,}402$. The `[all branching]` diagnostic used below means all unobserved states with more than one legal successor; these are the rows where the fitted model must generalise beyond directly observed station counts. On random-trip seed 42, $T = 20$, $|X_o| = 601$ and `maxiter` $= 1500$, the headline numbers are summarised in table 2.

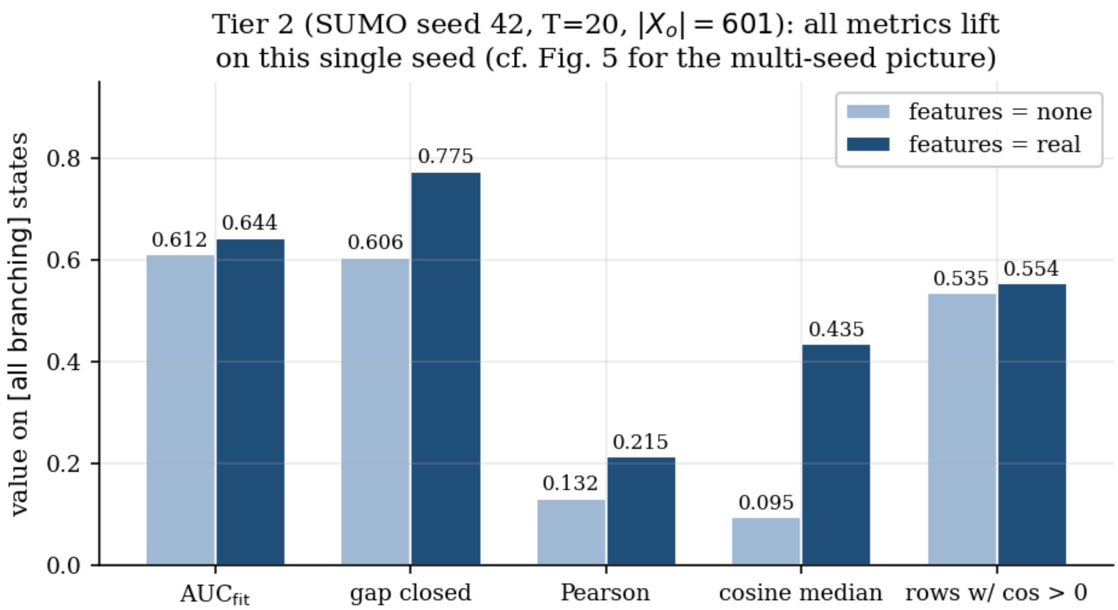

Figure 2: Tier 2 features lift on the Ingolstadt SUMO network, seed 42, $T = 20$. Every metric, including detection AUC, gap-closed, per-row Pearson, per-row cosine median and fraction of rows with positive cosine, moves positively under `features=real` (dark) vs. `features=none` (light). The Pearson +63% relative lift and the cosine median 4.6× lift on the `[all branching]` generalisation block are the strongest evidence that features improve chain recovery rather than aggregate-bias drift.

Table 2: Features lift across all metrics on the Ingolstadt SUMO network, seed=42 (fig. 2). `[all branching]` restricts the per-row diagnostic to states with out-deg > 1 that the inverter did not directly observe, namely the generalisation block.

metric (`[all branching]`)	`features=none`	`features=real`	Δ
$\mathrm{AUC}_{\mathrm{emp}}$ (full-obs. reference)	0.685	0.685	–
$\mathrm{AUC}_{\mathrm{fit}}$	0.612	**0.644**	+0.032
gap closed	60.6%	**77.5%**	+16.9 pp
stacked Pearson($\Delta_{\mathrm{fit}}, \Delta_{\mathrm{emp}}$)	+0.132	**+0.215**	+63% rel
per-row cosine median	+0.095	**+0.435**	4.6×
rows with cosine > 0	53.5%	55.4%	+1.9 pp

Two results are worth separating. First, the +0.032 AUC lift on this seed sits outside the single-instance variability measured in repeated SUMO baseline runs. As the multi-seed analysis of section 4.3.2 shows, however, this aggregate AUC lift does not survive replication across SUMO instances. The robust effect lies in the per-row chain-recovery improvement; the aggregate AUC does not replicate. Second, the stacked Pearson on `[all branching]` rises from +0.132 to +0.215, and the per-row cosine median from +0.095 to +0.435. These are the metrics that distinguish "the fit improved because the chains got more accurate per-row" from "the fit improved because the chains drifted in helpful opposite directions in aggregate". Both moved positively, supporting the stronger chain-recovery claim. Unlike the aggregate AUC, this per-row lift replicates across seeds (section 4.3.2).

The one metric that barely moves is the fraction of rows with positive cosine (only +1.9 pp on `[all branching]`, 58.6% at `[X_o branching]`). Features sharpen the contrast magnitude in the rows that were already approximately right, but do not flip the sign on the $\sim 45\%$ of branching states whose fitted contrast points the wrong way. This suggests the residual misspecification is structural to the 1-step softmax family. Section 4.3.3 tests that hypothesis.

4.3.2 Multi-seed validation: chain recovery is robust, detection AUC is not

To test whether the seed-42 lift replicates, the contrast-correlation diagnostic was run across five independent SUMO instances (`randomTrips` seeds $\in \{7, 23, 42, 101, 2024\}$, observation-set seed fixed at 13, $T = 20$, $|X_o| = 601$, `maxiter` $= 1500$) under both `features=none` and `features=real` (table 3, fig. 3). Because the inverse-MC loss is non-convex, the optimiser can settle into different local optima, or basins, on different SUMO instances. A basin here means a stable region of parameter space that L-BFGS-B reaches and then cannot improve beyond, even though another instance may reach a better region. The multistart audit in the Appendix tests whether random re-initialisation can escape the low basin.

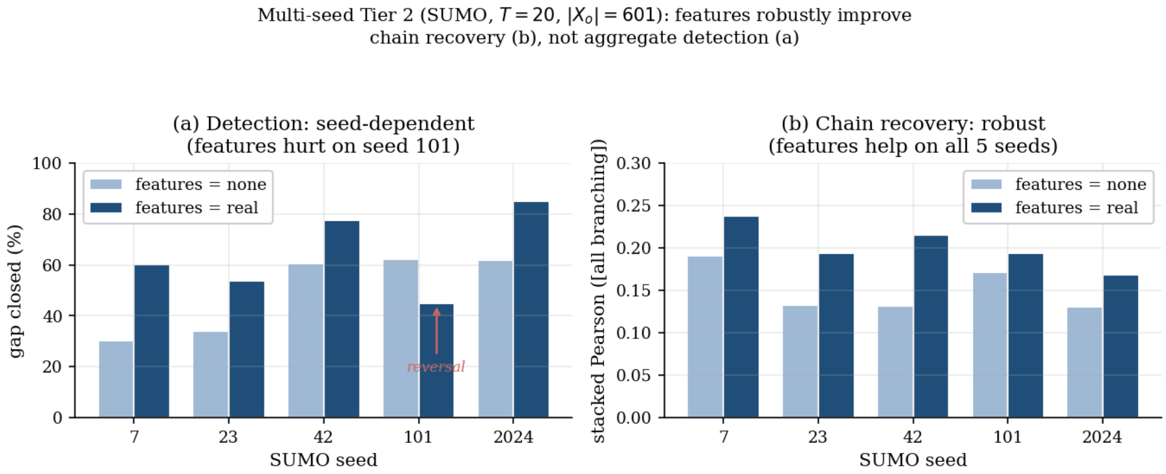

Figure 3: Multi-seed Tier 2 features sweep (five SUMO seeds, `[all branching]` block, $T = 20$). **(a)** The aggregate detection metric (gap-closed) lifts under `features=real` on four seeds but reverses on seed 101; the `features=none` bars also reproduce the bimodal basin structure ($\{7, 23\}$ low, $\{42, 101, 2024\}$ high). **(b)** The per-row chain-recovery metric (stacked Pearson) lifts on all five seeds. Features robustly improve chain recovery; aggregate detection remains seed-sensitive.

Table 3: Multi-seed features sweep on the Ingolstadt SUMO network (fig. 3), `[all branching]` block, $T = 20$, $|X_o| = 601$. The AUC lift reverses on seed 101; the Pearson lift is positive on all five seeds. Mean Δ rows are `real` − `none` with sample standard deviation.

	gap closed		$\mathrm{AUC}_{\mathrm{fit}}$		Pearson	
seed	none	real	none	real	none	real
7	30.3%	60.3%	0.544	0.588	+0.191	+0.238
23	34.2%	53.8%	0.554	0.585	+0.133	+0.194
42	60.6%	77.5%	0.612	0.644	+0.132	+0.215
101	62.1%	**45.0%**	0.581	**0.559**	+0.171	+0.194
2024	61.8%	85.1%	0.585	0.618	+0.131	+0.168
mean Δ	–		$+0.024 \pm 0.026$		$\mathbf{+0.050 \pm 0.023}$	

Three findings. First, the aggregate detection lift is not statistically robust: the paired $\mathrm{AUC}_{\mathrm{fit}}$ lift averages $+0.024 \pm 0.026$ across the five seeds, positive on four but reversing on seed 101. The test is paired because each SUMO seed is evaluated twice, once with `features=none` and once with `features=real`, so the statistic tests whether the within-seed AUC differences have a non-zero mean (paired t-test: $t \approx 2.0$, $p \approx 0.11 > 0.05$). The single-seed +0.032 headline of section 4.3.1 therefore reflects a favourable draw rather than a replicable property of the aggregate AUC.

Second, the per-row chain-recovery metrics are robust. The stacked Pearson lift averages $+0.050 \pm 0.023$ and is positive on all five seeds ($t \approx 4.9$, $p \approx 0.008$); the per-row cosine median lifts on all five as well (mean $\approx +0.25$). Seed 101 is the clearest warning that AUC and chain quality can move in different directions: features hurt its detection AUC ($0.581 \to 0.559$) while improving its chain recovery (cosine median $0.062 \to 0.475$). The features lever therefore improves **chain recovery** rather than aggregate detection.

Third, the `features=none` fit reproduces the bimodal gap-closure structure observed in repeated SUMO runs: seeds $\{7, 23\}$ cluster low (30–34%) and seeds $\{42, 101, 2024\}$ cluster high (61–62%), with nothing in between. The same seeds fall in the same clusters as the earlier run, indicating that the low/high split is a property of the loss landscapes on these instances rather than measurement noise. The `features=none` $\mathrm{AUC}_{\mathrm{fit}}$ across seeds is 0.575 ± 0.027, consistent with the baseline variability. A follow-up multistart audit tested whether the low basin could be escaped by random re-initialisation; the detailed audit is reported in Section D in the Appendix, and it did not escape the low basin.

4.3.3 Higher-order memory diagnostic: a regularised 2-step empirical lifts the reference

A natural hypothesis for the still-flipped rows is that the 1-step Markov assumption is too forgetful. The fitted model sees only the current edge x_t, but drivers may also condition on the edge they just came from: "do not U-turn", "continue along the arterial", or "stay in this lane group". A 2-step model gives the next-edge distribution the context (x_{t-1}, x_t) instead of only x_t.

The check uses full-observation empirical scorers only. If a regularised 2-step scorer ranks the two SUMO regimes better than the 1-step reference, the data contain route-memory signal outside the current 1-step softmax family. Without such a lift, the flipped rows point more plausibly to optimisation, sparse observation, or feature misspecification.

The catch is sparsity. On the Ingolstadt network the raw context space (x_{t-1}, x_t) has roughly $|X|^2 \approx 5.8\mathrm{M}$ possible pairs, and most possible pairs are never seen. A raw 2-step empirical matrix would therefore overfit rare contexts. Two regularisation schemes are tested across the same five SUMO seeds used in section 4.3.2 (training/test split unchanged):

Dirichlet smoothing: shrink rare contexts toward uniform turns.

$$\widehat{P}_{2,\alpha}(y|x_p, x) = \frac{c(x_p, x, y) + \alpha}{c(x_p, x) + \alpha \cdot |\mathrm{adj}(x)|},$$

swept over $\alpha \in \{10^{-3}, 10^{-2}, 10^{-1}, 1, 10, 100\}$.

Interpolation backoff: shrink rare contexts toward the 1-step empirical.

$$\widehat{P}_{\mathrm{back}}(y|x_p, x) = \lambda \cdot \widehat{P}_2^{\mathrm{MLE}}(y|x_p, x) + (1 - \lambda) \cdot \widehat{P}_1(y|x),$$

where $\widehat{P}_2^{\mathrm{MLE}}$ is the unsmoothed 2-step MLE and $\widehat{P}_1$ is the 1-step empirical chain that elsewhere serves as the full-observation reference. Two λ modes are tested:

- fixed $\lambda \in \{0, 0.25, 0.5, 0.75, 0.9, 1.0\}$;
- count-conditional $\lambda = N(x_p, x)/(N(x_p, x) + \kappa)$ with $\kappa \in \{1, 10, 100, 1000\}$.

The count-conditional mode is the intuitive version: trust the 2-step estimate when a context has been seen often, and fall back smoothly to the 1-step empirical for rare contexts.

Table 4: 2-step empirical sweep on the Ingolstadt SUMO network across five SUMO seeds ($T = 20$, training/test split as in section 4.3.2, 1-step empirical baseline mean 0.652 ± 0.022 across these seeds). Each row reports the best configuration for that method. The single-seed row is a comparison point; the multi-seed rows are the load-bearing evidence.

method	config	$\text{AUC}_{\text{2step}}$	Δ	wins
single-seed Dirichlet, seed 42	$\alpha \to 0$	0.666	−0.019	0/1
Dirichlet, multi-seed	$\alpha = 10^{-3}$	0.673 ± 0.012	+0.021	4/5
fixed backoff, multi-seed	$\lambda = 0.9$	0.747 ± 0.012	+0.095	5/5
count backoff, multi-seed	$\boldsymbol{\kappa = 100}$	$\mathbf{0.752 \pm 0.015}$	**+0.100**	**5/5**

Three findings (table 4).

First, the single-seed Dirichlet result is misleading. On seed 42 it gives 0.666, below the 1-step reference of 0.685, but seed 42 is also the highest-reference seed in the sweep. There is less room to improve on that seed than on the others. More importantly, Dirichlet smoothing shrinks rare 2-step contexts toward uniform outgoing turns, which throws away useful 1-step routing information.

Second, backoff to the 1-step empirical is the right shrinkage target. At $\kappa = 100$, the count-conditional rule gives positive lifts on all five seeds, ranging from +0.090 to +0.110 (mean +0.100, standard deviation 0.009). On seed 42 the effective λ across the 3,528 observed (x_p, x) contexts has median 0.21 and 10th–90th percentiles $[0.03, 0.81]$. In plain terms, the scorer mostly uses the trusted 1-step baseline and only lets the 2-step context dominate where the data support it.

Third, the conclusion is not sensitive to one hand-picked hyperparameter. The fixed-λ sweep gives lifts of +0.088 to +0.095 for $\lambda \in \{0.25, 0.5, 0.75, 0.9\}$, all positive across all five seeds. The count-conditional sweep gives +0.095 to +0.100 for $\kappa \in \{1, 10, 100, 1000\}$. The exact +0.100 maximum is mildly test-tuned, but the qualitative conclusion is stable: any reasonable backoff strength lifts AUC by about +0.09 to +0.10 across all five SUMO seeds.

Implication. A properly regularised 2-step empirical does lift the reference, decisively, across every SUMO seed tested: the regularised 2-step AUC reaches 0.752 ± 0.015 against the 1-step baseline of 0.652 ± 0.022, a +0.100 delta whose per-seed signs all point the same way. Two implications follow.

The per-row chain-recovery gap left by the 1-step features lever (section 4.3.1: +0.215 stacked Pearson on `[all branching]`, $\sim 45\%$ of rows still with wrong-direction contrast) is consistent with the 1-step softmax family being structurally unable to express (x_{t-1}, x_t)-conditional choice. This is the regime where a parametric 2-step Morimura extension is the right design lever, with the regularised empirical sitting roughly +0.10 AUC above the 1-step reference at $T = 20$ as the performance target it would need to approach.

No AUC_{fit} is reported for the regularised 2-step diagnostic. A parametric 2-step Morimura inverter is a non-trivial extension with a different state space, different gradient and different reference argument, and is left to future work (section 6.4). The empirical result above establishes that the signal exists and bounds the achievable 2-step inverter performance from above; what it does not do is settle whether the parametric inversion route can recover that bound under partial observation.

4.3.4 Trajectory-length sweep: lifting the reference and the detector together

The 0.685 reference is specific to the chosen test trajectory length $T = 20$. The log-likelihood-ratio score $\Lambda(\tau)$ is a sum of T independent log-ratio increments, so its variance scales as $\sim T$. Longer trajectories produce a tighter score distribution between the two regimes and a higher achievable AUC; the data are "the same", only the observation window is wider.

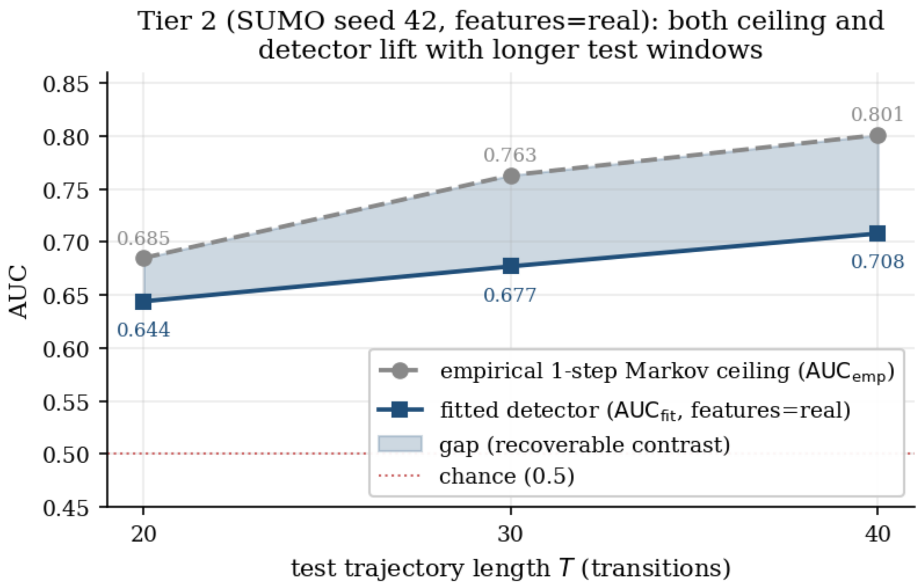

Figure 4: Tier 2 T-sweep on the Ingolstadt SUMO network, seed 42, `features=real`. The empirical 1-step Markov reference (grey dashed) rises with T as expected for a variance-shrinking score; the fitted detector (blue solid) tracks it with roughly constant gap. The shaded region is the recoverable contrast not yet captured by the fitter. The fitted chain itself is T-independent (identical final losses across runs); only the trajectory-level scoring window changes.

Table 5: T-sweep on the Ingolstadt SUMO network, seed=42, `features=real` (fig. 4). The fit itself is T-independent: only the test scoring window changes, as confirmed by the identical final-loss values 836.4/529.4 across all three runs.

T	AUC_{emp}	AUC_{fit}	gap closed	Pearson (all-br)
20	0.685	0.644	77.5%	+0.215
30	0.763	0.677	67.4%	+0.220
40	**0.801**	**0.708**	69.0%	+0.216

Two findings worth recording. First, the empirical reference rises from 0.685 to 0.801 between $T = 20$ and $T = 40$, confirming that the reference at $T = 20$ reflected the truncation window rather than a fundamental data limit. Second, the per-row Pearson is constant across T at $\sim +0.21$. T is purely a scoring-window lever: it lets the trajectory-level score discriminate more sharply, but it does not change the chain that was fit. The fitted chain captures the same $\sim 21\%$ of the true per-row contrast regardless of how long each test trip is observed.

The strongest single-seed SUMO headline is therefore AUC $= 0.708$ at $T = 40$, 69% of the gap closed against the full-observation empirical 1-step reference of 0.801, with a stable per-row chain Pearson of $+0.216$ on `[all branching]`.

A separate simulator stress test varied sat-navigation adoption and demand independently before the rush-vs-off-peak proxy was used in Tier 3. The full $\rho \times d$ decomposition is reported in Section E in the Appendix. In brief, under SUMO's static unaided-driver baseline, changing demand alone produced a near-random empirical AUC (0.493 ± 0.018), while adding full sat-navigation at peak demand produced a large signal (0.780 ± 0.013). This supports the rush/off-peak proxy only as a weak simulator audit: real commuters can remember recurring congestion, so the real-world proxy may still mix human experience with algorithmic guidance.

4.3.5 The $f + g$ extension at scale: a refined diagnosis

Morimura et al.'s estimator can use two kinds of sparse evidence. The f term says how much traffic is seen at each observed station; the g term says how often a trajectory starting at one observed station later reaches another. Intuitively, $f + g$ should be stronger than f alone because it adds coarse path-connectivity information beyond station busyness. The practical question is whether those extra hitting-rate measurements are dense and cheap enough to help at city scale.

The optional $f + g$ loss was audited because the initial SUMO-scale run at $|X_o| = 601$ did not terminate after 9.5 hours, but the diagnosis is more specific than simple infeasibility. A finite-difference-verified sparse-LU implementation confirms the mathematics, yet naive sparse solves are still slower than dense at full SUMO scale because the wide gradient solve dominates. On smaller Xuancheng subgraphs the solver is fast enough, but a different problem appears: most observed station pairs are never hit, so their empirical g values sit at the same small floor. Intermediate γ then lets this low-information g term overwhelm the useful count signal f; only high-γ settings close to the f-only end recover the empirical reference. The practical detector in this dissertation therefore remains f-only, and the solver profiling plus γ sweep are moved to Section B in the Appendix.

4.4 Interpretation

Tier 2 is the positive validation tier, but only under the controlled conditions that make its labels meaningful. With ground-truth rerouting labels and matched OD pairs, the fitted filter discriminates sat-navigation from static unguided routing, reaching AUC $= 0.708$ at $T = 40$ while closing 69% of the gap to the full-observation empirical 1-step reference. The feature-augmented score is useful primarily as a chain-recovery lever: it improves per-row Pearson/cosine consistently across five seeds, but its aggregate AUC lift is not robust.

The tier also identifies the model-class boundary. A regularised empirical 2-step reference lifts AUC by about $+0.09$–$+0.10$ on every SUMO seed, showing that higher-order route memory is a real signal in the controlled data and that a parametric 2-step inverse-MC model is the next justified detector design. By contrast, the $f + g$ extension is not ready at city scale under the current solver: subgraph tests show that it can help when γ is tuned near the f-only end, but full-network runtime and hitting-rate sparsity keep the deployed Tier 2 detector f-only. The defensible claim from this tier is therefore specific: the current 1-step partial-observation filter works as a controlled labelled detector, and its diagnostics correctly expose where richer model structure is needed.

5 Tier 3: Real-world stress test of the filter under OD confounding

5.1 Motivation

This tier stress-tests the partial-observation routing-influence filter (defined in section 2.3) against a real-world urban automatic vehicle identification (AVI) dataset from Xuancheng, China [22], with the OD-rebalancing audit (section 2.4) activated. The estimator, the scoring rule, and the per-row diagnostic are reused unchanged from Tier 2; what changes are the loader, the regime definition, and the OD-control logic. Because the real-world AVI datasets available for this study do not provide trip-level sat-navigation-use or route-guidance-compliance labels, this tier cannot directly validate sat-navigation detection. It is a real-world confounding audit that asks whether a plausible behavioural proxy (rush-hour vs. off-peak travel) contains a within-OD route-choice signal once the population-mix confound is controlled, and whether the filter survives that confounding. As the results below show, under the tested proxy the within-OD contrast is not recovered. The absence of a ground-truth label is the defining methodological constraint of this tier; the much shorter legalised trip windows are the main measurement constraint. Xuancheng sub-trajectories average only 8.2 edges, compared with 35.7 edges for the SUMO routes, so each real-world trajectory carries less likelihood-ratio evidence. The filter's OD-rebalancing and multistart audits exist precisely so that these constraints can be reported directly and kept separate from any spurious AUC.

5.2 Method

5.2.1 Dataset choice

Three candidate datasets were evaluated. The Padua AVI deposit [23] releases only aggregated 10-minute flow statistics and pairwise travel times over 40 junctions; per-vehicle trajectories are stripped at the source pipeline for privacy and cannot be reconstructed downstream. The Wang *et al.* Xuancheng deposit [24] explicitly restricts raw trajectory access behind a supervised-interactive interface. The Ma et al. [22] Xuancheng release, by contrast, publishes joinable entry/exit tables with anonymised vehicle identifiers from which complete per-vehicle trajectories can be reconstructed downstream; this dataset is used here. table 6 summarises its scale.

Table 6: Xuancheng AVI dataset [22] scale, compared with the Tier 2 Ingolstadt SUMO network.

property	Ingolstadt SUMO (Tier 2)	Xuancheng (Tier 3)
state space n	2,405 edges	1,733 edges (SCC subset)
trips per day	$\sim$ 28,000 per regime	$\sim$ 327,000
mean trajectory length	35.7 edges	15.1 edges (raw)
total trips (full pool)	$\sim$ 56,000	$\sim$ 10.5M (Apr 2023)
ground-truth regime label	yes (`rerouting` flag)	no (proxy required)

5.2.2 Verification: route legality and time-of-day distribution

Two non-obvious data-quality issues affect how the loader processes trips, each of which was identified before any fitting was attempted:

Route legality. Of all consecutive edge pairs (x_t, x_{t+1}) across the trip routes, only 78.4% satisfy the SUMO net's lane-connection definition $x_{t+1} \in \mathrm{adj}(x_t)$. The remaining 21.6% split as $\sim$ 1.9% junction-adjacent-but-disconnected (a definitional gap in the converted SUMO net) and $\sim$ 19.7% inter-junction "teleports" attributable to the source pipeline's Dijkstra-repair step that fills in missing AVI passages on CityFlow's connectivity graph instead of the SUMO net's. The loader handles this by splitting trips at illegal boundaries into legal-only sub-trajectories instead of dropping the trip wholesale: a 15-edge raw trip with one illegal jump becomes two sub-trajectories. This is the conservative choice because the inverse-MC scorer assigns likelihood only to legal successors in $\mathrm{adj}(x)$; treating source-pipeline repair jumps as real road transitions would contaminate the fitted chain with edges that do not exist in the analysed network. The cost is that Tier 3 is a test on legalised sub-trajectories rather than raw AVI trips. Mean sub-trajectory length is 8.2 edges, and the $T = 10$ experiments therefore draw their test set from the subset of sub-trajectories with at least 11 edges, selecting the longer and cleaner fragments of the reconstructed data.

Time-of-day distribution. Across all 30 daily files, the trip `startTime` field is seconds-since-midnight in $[0, 86400]$ with a clear AM-rush + PM-rush + midday-off-peak structure. This field is used to construct a rush-vs-off-peak regime split with windows rush $= [25{,}200, 32{,}400] \cup [61{,}200, 68{,}400]$ (7–9 AM and 5–7 PM) and offpeak $= [36{,}000, 57{,}600]$ (10 AM–4 PM); trips outside either window are discarded. Because the legalised Xuancheng sub-trajectories are much shorter than the SUMO routes, the Tier 3 tests use $T = 10$. This is shorter than every main SUMO scoring window ($T = 20$–40), and the Tier 2 sweep showed that AUC increases with longer T; the shorter Xuancheng window can therefore contribute to low test AUC even before any model-fitting error is considered. Three anomalous days are flagged for exclusion or special handling: Apr 5 (Tomb-Sweeping holiday, 264k trips, demand outlier), Apr 28–30 (Labour Day period, $\sim$ 366–426k trips), Apr 10 (`startTime` max $\approx$ 165k, suggesting a multi-day concatenation), and Apr 8/9 (identical trip counts of 324,441 in both files, treated as a quality-control flag).

5.2.3 The fundamental real-world gap: no ground-truth sat-nav label

Tier 2 used `-device.rerouting.probability` as ground truth: every vehicle either does or does not use sat-navigation, and the regime split is exact by construction. The real-world AVI datasets available for this study do not provide equivalent trip-level labels. The closest available Xuancheng proxy is therefore the rush-vs-off-peak time-of-day contrast: if external rerouting influence is stronger under congestion, rush-hour routes should differ from off-peak routes after OD mix is controlled.

This proxy is weaker than the SUMO contrast in two ways. First, rush and off-peak trips have different populations (commuters vs. discretionary trips), so any chain difference reflects both the routing-choice effect of interest and the nuisance OD-mix effect. Second, the actual sat-navigation adoption rate in Xuancheng is presumably much lower than the 100% used in SUMO, so even the routing-choice signal is diluted by the unguided fraction.

The OD-mix confound is addressed directly via the OD-rebalancing extension of section 2.4. The lower-adoption dilution cannot be addressed from observation alone, so the resulting real-world AUC is reported as a proxy-conditioned sensitivity stress test.

A further concern is that rush-vs-off-peak combines live route guidance with ordinary congestion adaptation. Tier 2 partly bounds this ambiguity: in the SUMO $\rho \times d$ decomposition (Section E in the Appendix), changing demand while holding sat-navigation adoption at $\rho = 0$ gives chance-level empirical AUC, while changing ρ at fixed peak demand gives a strong signal. That supports the proxy inside the simulator, but it does not settle the real-world case. SUMO's unaided drivers follow precomputed shortest paths, whereas real commuters may avoid recurring bottlenecks from experience. The Xuancheng proxy therefore serves as a behavioural stress test rather than labelled sat-navigation validation.

5.3 Results

5.3.1 Rush-vs-off-peak results

Three configurations are reported on the rush-vs-off-peak contrast: single-day, 5-day pooled, and 5-day pooled with $K = 8$ OD-rebalancing. Here K is the number of spatial zones used to coarsen origins and destinations before matching trip counts; $K = 8$ means trips are balanced across the shared cells of an 8-zone origin–destination grid.

The numerical evidence is split into two rush-vs-off-peak views. Table 7 shows the headline configurations, and table 8 isolates the OD-zone granularity sweep for the same proxy.

Table 7: Rush-vs-off-peak headline configurations on Xuancheng. Fitted columns report `features=none`/`features=real`; recovery diagnostics are Pearson and median cosine on `[all branching]` states.

configuration	$\mathrm{AUC}_{\mathrm{emp}}$	$\mathrm{AUC}_{\mathrm{fit}}$	recovery diagnostic
Apr 17 only, $T = 10$	0.556	0.498/0.508	Pearson $+0.139/+0.048$; cos $+0.341/+0.146$
Apr 17–21 pooled, $T = 10$	0.573	0.524/0.536	Pearson $+0.096/+0.072$; cos $+0.267/+0.167$
pooled + OD-match $K = 8$	0.518	0.476/0.530	Pearson $+0.104/+0.077$; cos $+0.332/+0.263$

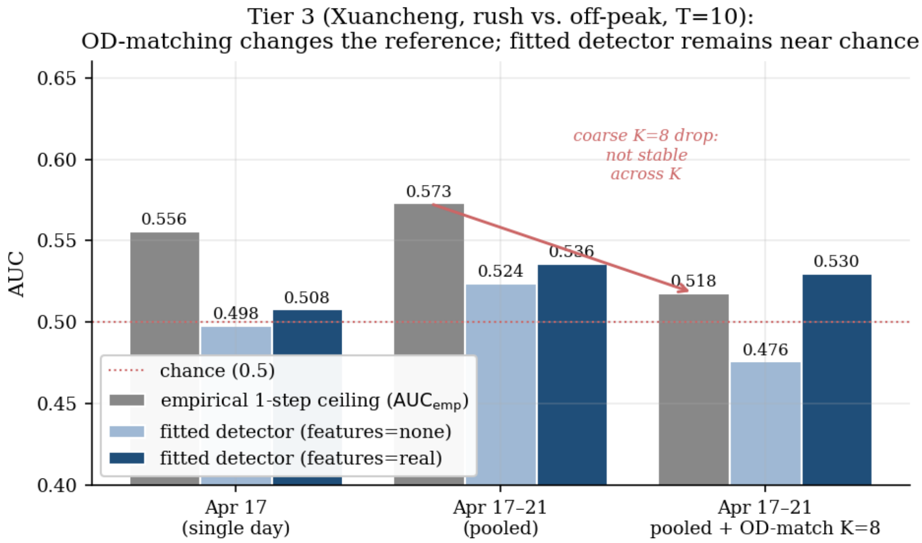

Figure 5: Tier 3 Xuancheng rush-vs-off-peak detection AUC across three configurations. The full-observation empirical 1-step reference (grey) is 0.556 on a single day, 0.573 when 5 weekdays are pooled, and 0.518 once the pools are OD-matched at $K = 8$ spatial zones. The apparent crash at $K = 8$ is not stable across zone granularity: a sweep over $K \in \{8, 16, 32, 64\}$ (table 8) shows the matched reference returning to $\sim$0.55–0.58 for $K \geq 16$. The $K = 8$ drop is therefore not treated as a fixed OD-mix fraction. Fitted detectors (blue bars) sit at or below chance in all three configurations.

Table 8: Rush-vs-off-peak OD-zone granularity sweep (`features=none`). Brackets give stratified vehicle-level bootstrap 95% confidence intervals.

OD control	shared OD cells	$\mathrm{AUC}_{\mathrm{emp}}$	$\mathrm{AUC}_{\mathrm{fit}}$
unmatched	–	0.573	0.524
$K = 8$	55	$0.518\ [0.471, 0.567]$	$0.476\ [0.429, 0.524]$
$K = 16$	157	$0.562\ [0.516, 0.605]$	$0.486\ [0.439, 0.530]$
$K = 32$	520	$0.553\ [0.507, 0.602]$	$0.513\ [0.467, 0.560]$
$K = 64$	1,448	$0.575\ [0.530, 0.622]$	$0.490\ [0.441, 0.535]$

The OD-matched $K = 8$ row matches 517,477 trips per regime across 55 shared OD cells. It lowers the full-observation reference from 0.573 to 0.518, but the granularity sweep shows that this drop is specific to coarse zoning: for $K \geq 16$, the matched empirical reference returns to about 0.55–0.58. The stable result concerns detector recovery: across every tested K, the fitted AUC remains at chance (0.476–0.513, with bootstrap intervals crossing 0.50), even where the full-observation chain weakly separates rush from off-peak routes.

The feature-augmented row needs separate handling. Its matched `features=real` AUC of 0.530 exceeds the empirical reference (0.518), indicating a fit problem rather than a better detector. The diagnostic is consistent with that reading: the rush-chain loss is 1397 under `features=real`, compared with 4.058 under `features=none`, and the per-row Pearson is lower with features (+0.077 vs. +0.104 on `[all branching]` states). The static road attributes also add parameters while the Xuancheng proxy is sparse and only indirectly related to route guidance. The multistart check in section 5.4.1 resolves the anomaly: once the high-loss basin is avoided, `features=real` returns to chance-level AUC and does not improve the real-world recovery.

5.3.2 Stronger-proxy stress test: Labour Day versus normal days

The rush-vs-off-peak proxy may simply be too weak. To test whether the detector works when the real-world behavioural contrast is stronger, the same pipeline was run on a holiday split suggested by the dataset audit: Labour Day period trips (Apr 28–30) against the same normal midweek pool (Apr 11–14 and Apr 17–21). A broader holiday pool that also included Tomb-Sweeping Apr 5 was first tested but gave empirical AUCs near chance at every OD granularity, consistent with pooling low-demand and high-demand holiday regimes that partially cancel. Labour-only is the more informative stress test.

Table 9: Labour-Day-versus-normal stress test (`features=none`). Brackets give stratified vehicle-level bootstrap 95% confidence intervals where available.

OD control	shared OD cells	$\mathrm{AUC}_{\mathrm{emp}}$	$\mathrm{AUC}_{\mathrm{fit}}$
unmatched	–	0.616 $[0.569, 0.661]$	0.520 $[0.472, 0.565]$
$K = 8$	57	0.541 $[0.494, 0.587]$	0.508 $[0.462, 0.554]$
$K = 16$	164	0.508 $[0.463, 0.553]$	0.524 $[0.478, 0.569]$
$K = 32$	556	0.529 $[0.481, 0.574]$	0.524 $[0.477, 0.568]$
$K = 64$	1,625	**0.567** $[0.521, 0.610]$	0.521 $[0.477, 0.568]$
$K = 64$, multistart	1,625	0.567	**0.502**

This addresses the concern that the real-world tier is negative only because rush-vs-off-peak is too weak. Table 9 and fig. 6 show that Labour Day produces a clearer route-choice contrast: the unmatched empirical reference is 0.616, and after fine OD matching at $K = 64$ it remains above chance at 0.567 (CI $[0.521, 0.610]$). The fitted chain, however, reaches only 0.521 under the zero-init fit and drops to 0.502 under validation-selected multistart. The per-row diagnostic is also weak (Pearson +0.066 on all branching states). Thus, even when Xuancheng contains a measurable within-OD 1-step contrast, the current partial-observation parametric recovery does not recover it robustly.

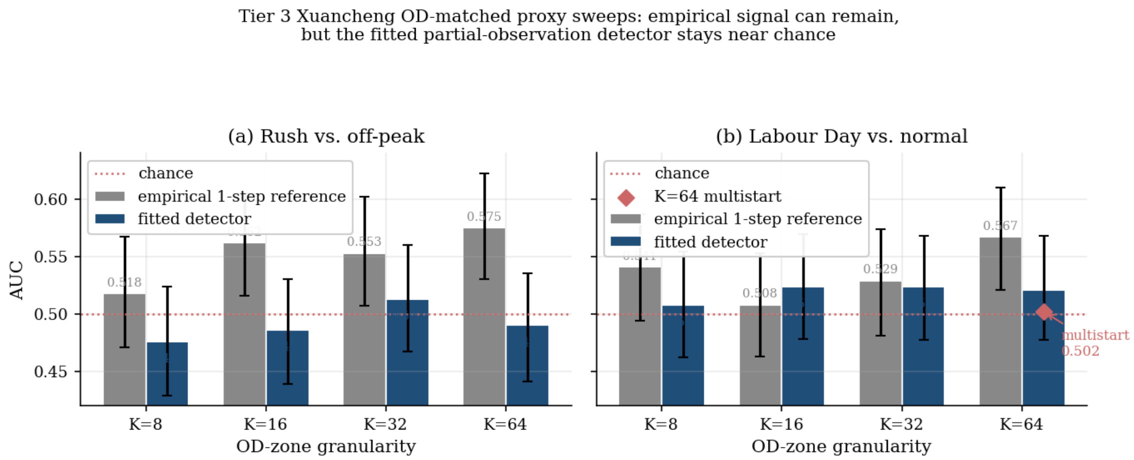

Figure 6: Tier 3 OD-matched proxy sweeps. **(a)** In the rush-vs-off-peak proxy, the empirical reference returns above chance for $K \geq 16$, but the fitted detector stays near chance. **(b)** Labour Day creates the clearer empirical contrast at $K = 64$, yet the fitted detector remains weak and the validation-selected multistart result drops to 0.502. Error bars are bootstrap 95% confidence intervals for the shown AUCs.

A final check asks whether route memory carries the real-data signal that first-order turning probabilities miss. It reuses the fitting-free empirical 2-step backoff diagnostic from Tier 2. The question is whether the mechanism that helped SUMO – next-edge choice depending on both the current and previous edge – is also visible in the real proxy. The diagnostic was run on the same Labour $K = 64$ split, using the interpolation-backoff family that produced the Tier 2 SUMO lift.

The result is negative. The 1-step empirical reference remains 0.567 (CI $[0.521, 0.610]$), while every non-zero 2-step interpolation weight lowers AUC (fixed $\lambda = 0.5$: 0.558; count-conditional $\kappa = 100$: 0.542). The best setting is $\lambda = 0$, the original 1-step chain. Unlike the labelled SUMO experiment (section 4.3.3), Xuancheng Labour-vs-normal gives no preliminary evidence that simple second-order route memory is the missing real-world lever. The remaining gap is more likely due to partial-observation identifiability, proxy noise, unobserved covariates, or OD/time heterogeneity.

5.4 Interpretation

The Xuancheng result is a negative validation result for the current partial-observation detector under behavioural proxies. It should not be read as evidence that the city has no route-choice signal. Fine OD matching leaves a weak full-observation rush-vs-off-peak contrast in several settings, and the Labour-Day split makes that contrast clearer. The fitted 1-step inverse-MC detector still stays at chance after OD control and after multistart validation selection. This points to partial-observation identifiability under a noisy proxy: only 433 of 1,733 states anchor the observed count ratios, while most branching transitions must be inferred through a parametric softmax from sparse observations.

The feature and route-memory audits narrow the failure mode. Static road features do not help because they increase the fitted parameterisation in a sparse, proxy-labelled setting and, on Xuancheng, lead to an unstable high-loss basin unless multistart is used;

the validation-selected feature fit returns to chance. A fitting-free 2-step empirical backoff on the Labour-Day split also fails to improve the full-observation reference. The remaining explanation combines sparse observation, proxy noise, OD/time heterogeneity and limited partial-observation identifiability.

5.4.1 Multistart on `features=real`: the ceiling violation is an optimiser artefact

The one anomaly left open above is the OD-matched $\mathrm{AUC}_{\mathrm{fit}} = 0.530 > \mathrm{AUC}_{\mathrm{emp}} = 0.518$ under `features=real`, with its tell-tale 1397 rush-chain loss. This is suspicious rather than encouraging: the empirical reference is built from the full-observation transition matrix, while the inverse-MC detector sees only sparse station counts. A fitted partial-observation detector can exceed that reference by finite-sample noise, but a large excess combined with a very high training loss is more naturally read as an optimiser artefact. The multistart variant was run (`score_seed_big_multistart.py`, $K = 5$ random restarts $\theta_0 \sim \mathcal{N}(0, I)$ per chain, validation-selected by held-out log-likelihood) on the exact OD-matched `features=real` configuration (d = 6,136, T = 10, $|X_o|$ = 433, 517,177 train transitions per regime).

The result is decisive and opposite to the SUMO basin check of section D. On Xuancheng, multistart does escape the high-loss basin: the rush (intrinsic) chain loss drops from the zero-init baseline of 1397 to 9.186 at the validation-selected restart (restart losses $\{1413, 1844, 273.4, 9.186, 1792\}$, so one restart of five found the good basin). But the well-fit chains then give $\mathrm{AUC}_{\mathrm{fit}} = 0.483$, below chance, against the baseline's spurious 0.530. The per-init paired AUCs ($\{0.504, 0.492, 0.497, 0.494, 0.491\}$, mean 0.496) cluster tightly at chance regardless of which basin each restart lands in. The optimiser instability is real and severe; escaping it still does not recover a usable behavioural proxy signal.

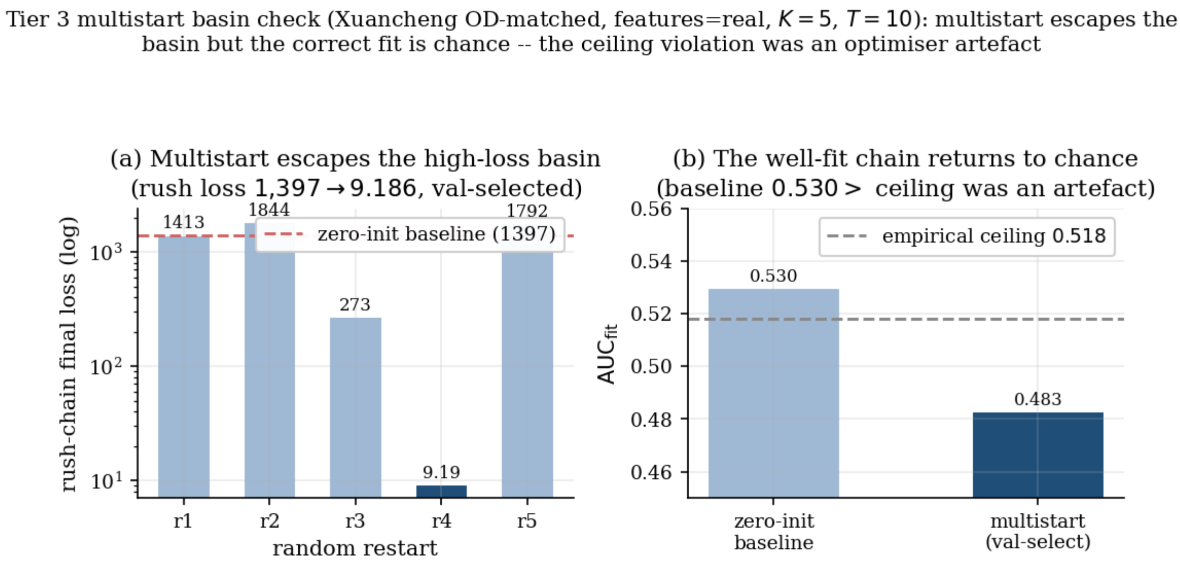

Figure 7: Xuancheng OD-matched `features=real` multistart ($K = 5$). **(a)** The rush-chain training loss collapses from the zero-init baseline (1,397) to the validation-selected restart (9.186): random re-initialisation escapes the high-loss basin, unlike the SUMO case (section D). **(b)** Despite the proper fit, $\mathrm{AUC}_{\mathrm{fit}}$ falls from the baseline's spurious 0.530 (which exceeded the empirical ceiling 0.518) to 0.483 – chance. The ceiling violation was an optimiser artefact; under correct chain recovery the fitted detector returns to chance.

The multistart result removes the apparent ceiling violation without changing the scientific conclusion. The high-loss `features=real` fit produced the anomalous 0.530 AUC; the validation-selected restart fits the training objective much better and returns chance-level test AUC. The SUMO and Xuancheng basin checks therefore play different roles: SUMO shows that the low basin is stable under re-initialisation, while Xuancheng shows that escaping a bad basin still does not recover a usable behavioural-proxy signal.

5.4.2 Bottom line for Tier 3

The pipeline runs cleanly on real-world data: the gradient checks, regime-split machinery and contrast diagnostics all port to Xuancheng. The result is also consistent across the two real-world proxies. In the rush-vs-off-peak split, the fitted detector stays at chance for $K \in \{8, 16, 32, 64\}$ even when the full-observation reference is weakly above chance. In the Labour-Day split, the reference is stronger (0.567 after fine OD matching), but the validation-selected fitted detector is still at chance (0.502). The controlled SUMO tier therefore validates the machinery under labelled conditions; Xuancheng shows that the same machinery cannot directly validate sat-navigation detection when only behavioural proxy labels are available.

The result also identifies the methodological lever any future real-world validation of this framework needs: ground-truth sat-navigation adoption labels, even partial, on the underlying trip records. Such labels would remove the proxy-label ambiguity and allow the controlled-comparison machinery of Tier 2 to be repeated on real trips, while OD balance, data quality and adoption heterogeneity would still need to be audited. Without those labels, the fitted 1-step detector cannot separate genuine within-OD routing differences from chance robustly on these behavioural proxies.

6 Discussion

6.1 Cross-tier comparison

The core inverse-MC scoring machinery works cleanly across all three tiers: the inverter, FD-checked gradient, contrast diagnostic, and feature plumbing operate unchanged on a synthetic $n = 100$ chain, a SUMO $n = 2{,}405$ network, and a real-world $n = 1{,}733$ Xuancheng network. What differs by tier is the loader, the regime definition, and the OD-control logic; the shared scoring core is selected by a single `-dataset {sumo,xuancheng}` command-line interface (CLI) flag. This is a useful generalisation result for an inverse-MC pipeline that the original paper validated only on synthetic data and one real-world demonstration.

What does not generalise as cleanly is the magnitude of the detection signal. Tier 1 reproduces the paper's RMAE within ~ 0.05 for $|X_o| \geq 10$. Tier 2 reaches AUC $= 0.708$ at $T = 40$ on the controlled SUMO comparison, $\sim 69\%$ of the gap to the full-observation empirical 1-step reference of 0.801, with per-row chain Pearson $\sim +0.21$. Tier 3 on Xuancheng, with no ground-truth sat-navigation label, gives chance-level fitted AUC on the rush-vs-off-peak behavioural proxy. A stronger Labour-Day proxy creates a significant fine-OD empirical signal ($\text{AUC}_{\text{emp}} = 0.567$), but multistart validation-selected fitting still falls to chance ($\text{AUC}_{\text{fit}} = 0.502$). The drop between Tier 2 and Tier 3 has two distinct contributions:

- The Xuancheng ceiling itself ($\text{AUC}_{\text{emp}} \sim 0.57$) is much lower than the SUMO ceiling (~ 0.69), reflecting that the behavioural rush-vs-off-peak proxy is a weaker contrast than the SUMO labelled comparison.
- Xuancheng is also scored on shorter legalised trajectories ($T = 10$), whereas the main SUMO detector starts at $T = 20$ and improves again at $T = 30$ and $T = 40$. Since the LLR score accumulates evidence over the trajectory, the shorter real-data window likely suppresses both the empirical reference and fitted AUC.
- The fitted-vs-reference gap is not merely a scaled-down version of the SUMO success. The pooled un-matched Xuancheng row gives a superficially moderate gap-closure number because the reference itself is weak, but once OD mix is controlled the fitted detector sits at chance across every tested K while the full-observation reference remains weakly above chance at finer granularities. Tier 3 is therefore a negative result for the current 1-step partial-observation filter under this behavioural proxy.
- The Labour-Day stress test rules out the strongest alternative reading: that a more intuitive real-world contrast would rescue the detector. Labour Day does yield a stronger empirical reference after fine OD matching, but the fitted chain remains near chance and multistart selection removes the small zero-init lift.
- A fitting-free 2-step backoff audit on the same Labour-Day split also fails to lift the empirical reference: the best setting is the degenerate $\lambda = 0$ backoff, equivalent to the original 1-step chain. The SUMO route-memory result therefore motivates a future model class, but it is not by itself an explanation of the Xuancheng failure.

Under controlled labelled simulation the method recovers a robust fraction of the per-row chain contrast. On a real-world behavioural proxy, the fitted 1-step detector remains near chance once population-mix is controlled.

Table 10 consolidates the three tiers on a common axis: the same inverter and diagnostic produce every row, and the trend in the empirical ceiling, from a noise-free synthetic recovery, through a labelled simulated comparison, to a behavioural real-world proxy, quantifies how much detectable signal each setting actually contains.

Table 10: Cross-tier synthesis. Tier 2 reports the representative Ingolstadt seed 42 at $T = 20$; Xuancheng rows report the weekday proxy and the Labour-Day stress test. Fitted and Pearson cells are `none`/`real`.

tier	dataset (n)	empirical ceiling	fit AUC	Pearson
1 synthetic	random graph (100)	–	–	–
2 SUMO	Ingolstadt network (2,405)	0.685	0.612/0.644	$+0.132/+0.215$
3 Xuancheng	AVI (1,733)	0.573	0.524/0.536	$+0.096/+0.072$
3 Xuancheng Labour	AVI (1,733)	0.567	$0.521 \to 0.502$	$+0.066$

6.2 Limits of the chosen approach

The main limitations are label availability, stationarity, and model capacity.

Proxy labels and stationarity. Tier 3 cannot observe sat-navigation use directly, so rush-vs-off-peak and Labour-Day splits are behavioural proxies. They mix route guidance, ordinary congestion adaptation, driver composition and OD mix. The estimator also compresses dynamic time windows into one stationary surrogate chain, which is useful for scoring but should not be interpreted as a literal invariant traffic distribution.

Controlled simulation versus real traffic. Tier 2 provides a labelled test bed, yet it still contains simulator-specific structure: at the selected demand the sat-navigation regime completes many more trips, and unaided SUMO drivers follow precomputed routes. The $\rho \times d$ audit in Section E in the Appendix shows that this simulator signal is mainly tied to the sat-navigation label, but a real experienced driver can adapt without live route guidance. This is why Tier 3 remains a proxy stress test until trip-level guidance labels are available.

Sparse observation, optimisation and feature capacity. Both SUMO and Xuancheng show that the inverse problem is under-constrained at city scale. Multi-seed SUMO runs make the feature result weaker as an AUC claim but robust as a per-row chain-recovery effect; Xuancheng multistart removes an apparent `features=real` ceiling violation and returns the fitted detector to chance. The $f + g$ extension adds another capacity route, but its city-scale implementation remains expensive because each iteration needs large linear solves. The current thesis therefore uses the tractable f-only detector and treats $f + g$ and higher-order dynamics as model-class extensions.

6.3 Social, environmental, technological implications

Transport-policy modelling. Navigation platforms can change the routes that appear in revealed-preference data. If those influenced routes are fed back into demand models as ordinary preferences, policy evaluation can confuse platform response with driver preference. The controlled SUMO result shows that inverse-MC filtering can recover part of this distinction when labels exist; the Xuancheng result shows that behavioural proxies alone are too weak for confident attribution.

Privacy and data access. The evidence needed for this task is awkward: trip-level trajectories are useful, but trip-level app-use labels are sensitive. Aggregate-only releases protect privacy but cannot support the detector tested here; joinable per-trip releases enable the method while increasing reidentification risk. A practical deployment would need privacy-preserving access to coarse guidance/compliance labels, or a trusted-analysis setting in which labels can be used without publishing raw identifiers.

Platform accountability. A reliable detector would let cities audit how much of observed route choice is shaped by guidance systems, without needing to inspect individual recommendations. This matters for congestion pricing, traffic-calming decisions and claims about network welfare. The negative Tier 3 result is part of that accountability story: it warns against treating rush-hour AVI contrasts as evidence of sat-navigation influence unless the proxy is supported by labels or a stronger behavioural model.

6.4 Future work

1. **Labelled real-world validation.** The highest-value next dataset would pair trajectories with coarse sat-navigation-use or route-guidance-compliance labels. Even partial labels would convert Tier 3 from a proxy stress test into a direct validation exercise.
2. **Robustness over stations, OD cells and cities.** The current fits condition on one observation-station set and a fixed OD match. A deployment study should repeat the inverse-MC fits over multiple X_o draws, resample OD cells, and replicate across more city/month pools.
3. **Richer behavioural state.** SUMO shows that second-order route memory can carry signal, while Xuancheng Labour does not show a simple empirical 2-step lift. A parametric extension should therefore test higher-order context, time-of-day conditioning and trip-level features as competing explanations, with the chain-recovery claim kept separate from any deployment-oriented classifier.
4. **Scalable $f + g$ optimisation.** The hitting-rate extension remains attractive because it uses more of Morimura's original evidence, but city-scale fitting needs factor sharing or a different optimiser. Sherman–Morrison updates for the $(I - \beta P_T)$ solves and natural-gradient or Levenberg–Marquardt steps are the most direct algorithmic candidates.
5. **Experienced-driver simulation.** The Tier 2 $\rho \times d$ decomposition should be repeated with unaided drivers who avoid known rush-hour bottlenecks. That counterfactual would make the simulator bound closer to the real Xuancheng ambiguity between human experience and algorithmic guidance.

7 Conclusion

This dissertation designed and audited a partial-observation routing-influence filter. The filter ingests a road graph, sparse regime-labelled station observations at X_o, and held-out trajectories; it fits one inverse-Markov-chain model per regime; then it scores each trajectory with a log-likelihood ratio. The underlying estimator is Morimura et al.'s inverse-MC method [1]; the contribution here is the two-regime detector built around it, the trajectory-level scoring rule, and the diagnostic suite used to decide whether a positive AUC is actually evidence of route-choice recovery.

What the evidence supports. Tier 1 validates the estimator implementation: on Morimura et al.'s synthetic recovery benchmark, the reproduced RMAE curve tracks the published proposed-method curve for $|X_o| \geq 10$, with analytic gradients finite-difference checked before downstream use. Tier 2 is the positive detector result. In labelled SUMO simulation, with shared OD pairs and a known rerouting flag, the filter reaches AUC $= 0.708$ at $T = 40$ against a full-observation empirical 1-step reference of 0.801. The feature-regularised variant does not robustly lift aggregate AUC across all seeds, but it does robustly improve per-row chain recovery. Tier 3 is the real-world stress test. On Xuancheng AVI data, where only behavioural proxy labels are available, the fitted 1-step detector remains near chance after OD control: rush-vs-off-peak does not recover at any tested OD granularity, and the stronger Labour-Day-vs-normal split falls to AUC $= 0.502$ under validation-selected multistart despite a weak full-observation empirical signal.

What the audits changed. Several headline-looking results became more precise once audited. The SUMO features result became a chain-recovery result rather than a robust detection-AUC result. The SUMO 2-step empirical backoff lifted AUC by about $+0.09$–$+0.10$ across all five seeds, showing that higher-order route memory is a real signal in the controlled data and a justified future model class. The same 2-step audit did not help on the Xuancheng Labour-Day split, so route memory is not the simple missing lever for that real-world proxy. The optional $f + g$ route is mathematically verified and useful on small subgraphs when γ is tuned near the f-only end, but naive city-scale sparse-LU is too slow and empirical hitting rates are too floor-dominated to use as the deployed full-network detector in this thesis.

Where the filter is reliable. The filter is reliable as a controlled labelled detector under the conditions tested in SUMO: matched OD pairs, known rerouting labels, and a finite-data empirical reference against which chain recovery can be audited. In that setting it recovers a substantial fraction of the available 1-step signal and exposes where the 1-step model class runs out of capacity.

Where it is not reliable. The filter does not support a positive real-world sat-navigation detection claim on Xuancheng. Without trip-level guidance/adoption labels, rush-vs-off-peak and Labour-Day splits remain behavioural proxies that mix route guidance, ordinary congestion adaptation, driver composition, OD mix, time heterogeneity and data-quality effects. Once those confounds are audited, the fitted partial-observation detector does not robustly recover the within-OD signal that remains in the full-observation reference.

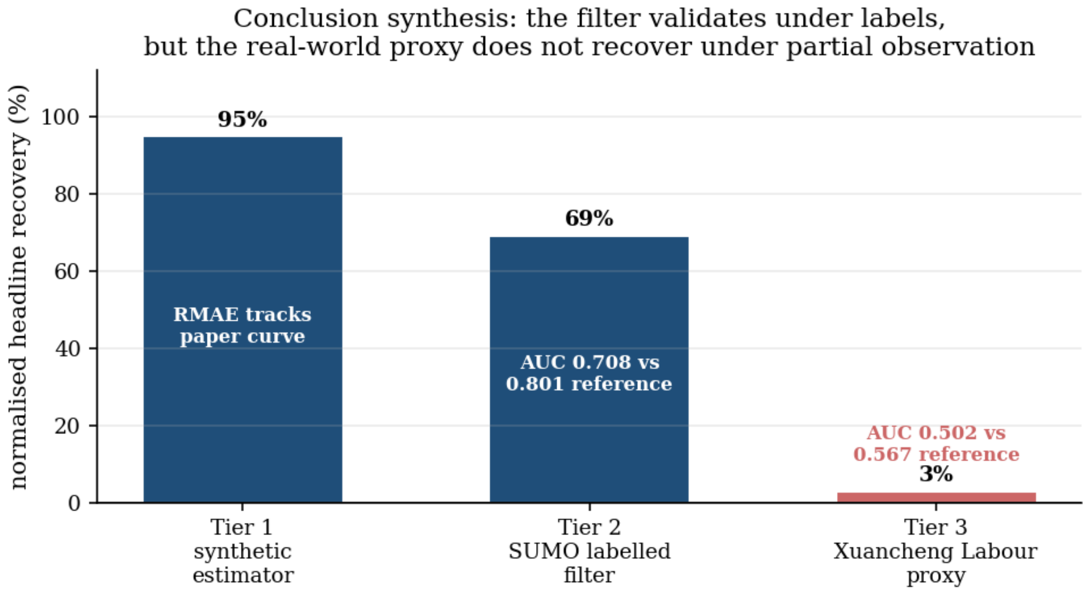

Figure 8: Cross-tier synthesis of the dissertation's headline results. The bars use a normalised summary, since the tiers use different metrics: Tier 1 reports benchmark-reproduction closeness, computed as the mean capped ratio between the visual paper RMAE and this reproduction's RMAE for $|X_o| \geq 10$; Tiers 2–3 report AUC gap closed, $(\mathrm{AUC}_{\mathrm{fit}} - 0.5)/(\mathrm{AUC}_{\mathrm{emp}} - 0.5)$. The central pattern is that the estimator reproduces its synthetic benchmark, the labelled SUMO filter recovers a substantial fraction of the empirical 1-step signal, and the real Xuancheng behavioural proxy does not recover under the fitted partial-observation model.

In sum, partial observation is the practical reason for the filter, not just the name of the estimator: many realistic transport datasets provide station counts, AVI detections or incomplete trajectory fragments rather than complete routes linked to app-use labels. The inverse-MC machinery is therefore useful as a sparse-evidence filter, but it is not the main bottleneck for real-world externally-influenced routing detection; validation data is the blocker. A future deployment needs at least partial trip-level sat-navigation-use or guidance-compliance labels, repeated station and OD sensitivity checks, and a richer model class where the controlled data show it is warranted. Until those labels exist, this framework is best read as a validated controlled-simulation filter and a real-world confounding audit rather than a finished real-world sat-navigation detector.

8 Project Management and Reflection

8.1 Project structure and milestone management

The project was scoped at the outset as a three-tier validation pipeline (synthetic, simulated, real-world) extending a single source paper [1]. This decomposition gave the work a useful order: first verify the estimator on the paper's own benchmark, then test the full detector in labelled simulation, then stress-test the same pipeline on real data where the labels are only proxies.

Milestone management was driven by a continuously updated `research_logbook.md`. Each experimental cycle recorded the hypothesis, setup, result, interpretation, and next action. The logbook became especially useful when results changed direction, because it made clear which claims were still supported by evidence and which were only working assumptions from an earlier stage.

Two practices worked well:

- **Audit-before-extend.** An early retraction of a headline number made it clear that every extension needed an upstream audit first. The gradient finite-difference check at $\sim 10^{-9}$ relative error confirmed the analytic gradient derivation before the feature sweep, and the contrast-correlation diagnostic showed that AUC alone was not enough to claim chain recovery.
- **Keeping decisions traceable.** Linking each reported number back to a script, seed, log entry and interpretation note made the late rewrite possible. It was particularly important for the $f+g$ diagnosis, the SUMO multistart check and the Labour-Day stress test, all of which changed how the results should be framed.

Two practices that did not work as well are worth recording explicitly:

- **Underestimating the compute scaling of the $f+g$ loss.** The first $f+g$ sweep at SUMO scale was projected at ~ 1 hour and killed at 9.5 hours. The retry at smaller $|X_o|$ and `maxiter` was projected at ~ 30 minutes and killed at 3 hours. Two iterations were burned before the structural diagnosis (the hitting-rate gradient is $O(|X_o| \cdot n^3)$ and the 84%-floor loss landscape is degenerate at this scale) was reached. Future sweeps at unfamiliar configurations should profile a single iteration before committing to a full run.
- **Late identification of the OD-mix confound.** The Tier-3 rush-vs-off-peak proxy was scoped before the methodological cost of the OD-mix confound was clearly understood. The OD-rebalancing extension, the methodologically correct fix, was added late in the project, and the matched-pool result it produced (section 5.4) materially changed the Tier-3 interpretation. Earlier scoping would have caught this; the lesson is that "proxy weakness" should be a structured pre-experiment audit item, not an ad-hoc discussion item.

8.2 Personal skill development

The main technical skills developed through the project were deriving and finite-difference-verifying analytic gradients (Eqs. 12, 15, 17–18 of the source paper); writing software with a clean separation between the inverter's mathematical core (in `morimura.py`, kept faithful to the paper) and the application-specific loaders that surround it (`sumo_to_phase3.py`, `real_data/load_xuancheng.py`); and maintaining a logbook that records both the successful and the unsuccessful experimental cycles in enough detail for later critical review.

The non-technical skill that most shaped the project's outcome was the disciplined treatment of negative results. The project did not produce the headline "real-world detector with a high AUC and clean chain recovery" that the SUMO Tier-2 result would have predicted. Instead it produced: a controlled-validation result (~ 0.71 AUC, 69% gap closure on labelled SUMO), a model-class diagnostic (the empirical 2-step detector improves the 1-step reference, strengthening the case for a parametric 2-step extension), and a real-world result (~ 0.53 AUC on a behavioural proxy without ground-truth labels) that quantifies the gap between simulated and real-world detectability. Each of these is a methodologically useful result, in part because they were framed accurately. The temptation late in the project to treat a stronger holiday proxy as a headline rescue had to be handled carefully rather than either embraced uncritically or dismissed. Running it with the same OD controls, bootstrap intervals, per-row diagnostics and multistart audit showed the useful version of the negative result: Labour Day has a real full-observation within-OD signal, but the fitted detector still falls to chance under validation-selected multistart. Accepting that result, rather than chasing a higher headline number, was an important project-management lesson.

References

[1] T. Morimura, T. Osogami, and T. Idé, "Solving inverse problem of Markov chain with partial observations," in *Advances in Neural Information Processing Systems 26 (NeurIPS)*, 2013, pp. 1655–1663, https://papers.neurips.cc/paper/5122-solving-inverse-problem-of-markov-chain-with-partial-observations.

[2] H. Yang and Q. Meng, "Modeling user adoption of advanced traveler information systems: Dynamic evolution and stationary equilibrium," *Transportation Research Part A: Policy and Practice*, vol. 35, no. 10, pp. 895–912, 2001, https://doi.org/10.1016/S0965-8564(00)00030-6.

[3] G. d. M. Ramos, T. Mai, W. Daamen, E. Frejinger, and S. P. Hoogendoorn, "Route choice behaviour and travel information in a congested network: Static and dynamic recursive models," *Transportation Research Part C: Emerging Technologies*, vol. 114, pp. 681–693, 2020, https://doi.org/10.1016/j.trc.2020.02.014.

[4] D. Acemoglu, A. Makhdoumi, A. Malekian, and A. Ozdaglar, "Informational Braess' paradox: The effect of information on traffic congestion," *Operations Research*, vol. 66, no. 4, pp. 893–917, 2018, https://doi.org/10.1287/opre.2017.1712.

[5] T. Cabannes, M. A. Sangiovanni-Vincentelli, A. Sundt, H. Signargout, E. Porter, V. Fighiera, J. Ugirumurera, and A. M. Bayen, "The impact of GPS-enabled shortest path routing on mobility: A game theoretic approach," in *Transportation Research Board 97th Annual Meeting*, 2018, https://trid.trb.org/View/1495267.

[6] W. Krichene, B. Drighes, and A. M. Bayen, "On the convergence of no-regret learning in selfish routing," in *Proceedings of the 31st International Conference on Machine Learning (ICML)*, ser. Proceedings of Machine Learning Research, vol. 32, no. 2, 2014, pp. 163–171, https://proceedings.mlr.press/v32/krichene14.html.

[7] J. P. Epperlein, S. Zhuk, and R. Shorten, "Recovering Markov models from closed-loop data," *Automatica*, vol. 103, pp. 116–125, 2019, https://doi.org/10.1016/j.automatica.2019.01.022.

[8] Y. Gu, E. Crisostomi, M. Liu, and R. Shorten, "Identification of new patterns in urban traffic flows," in *2018 IEEE Conference on Control Technology and Applications (CCTA)*, 2018, pp. 626–631, https://doi.org/10.1109/CCTA.2018.8511490.

[9] A. A. Kurzhanskiy, "A methodology for estimating vehicle route choice from sparse flow measurements in a traffic network," *Mathematics*, vol. 10, no. 3, p. 527, 2022, https://doi.org/10.3390/math10030527.

[10] J. Rust, "Optimal replacement of GMC bus engines: An empirical model of Harold Zurcher," *Econometrica*, vol. 55, no. 5, pp. 999–1033, 1987, https://doi.org/10.2307/1911259.

[11] M. Ben-Akiva and S. R. Lerman, *Discrete Choice Analysis: Theory and Application to Travel Demand*. MIT Press, 1985, https://mitpress.mit.edu/9780262536400/discrete-choice-analysis/.

[12] E. Frejinger and M. Bierlaire, "Capturing correlation with subnetworks in route choice models," *Transportation Research Part B: Methodological*, vol. 41, no. 3, pp. 363–378, 2007, https://doi.org/10.1016/j.trb.2006.06.003.

[13] M. Fosgerau, E. Frejinger, and A. Karlström, "A link based network route choice model with unrestricted choice set," *Transportation Research Part B: Methodological*, vol. 56, pp. 70–80, 2013, https://doi.org/10.1016/j.trb.2013.07.012.

[14] T. Mai, E. Frejinger, and M. Fosgerau, "A nested recursive logit model for route choice analysis," *Transportation Research Part B: Methodological*, vol. 75, pp. 100–112, 2015, https://doi.org/10.1016/j.trb.2015.03.015.

[15] E. Crisostomi, S. Kirkland, and R. Shorten, "A google-like model of road network dynamics and its application to regulation and control," *International Journal of Control*, vol. 84, no. 3, pp. 633–651, 2011, https://doi.org/10.1080/00207179.2011.568005.

[16] J.-G. Lee, J. Han, and X. Li, "Trajectory outlier detection: A partition-and-detect framework," in *Proceedings of the 24th IEEE International Conference on Data Engineering (ICDE)*. IEEE, 2008, pp. 140–149, https://doi.org/10.1109/ICDE.2008.4497422.

[17] T. P. van Oijen, W. Daamen, and S. P. Hoogendoorn, "Estimation of a recursive link-based logit model and link flows in a sensor equipped network," *Transportation Research Part B: Methodological*, vol. 140, pp. 262–281, 2020, https://doi.org/10.1016/j.trb.2020.08.003.

[18] T. Mai, E. Frejinger, M. Zimmermann, and F. Bastin, "Estimation of recursive route choice models with incomplete trip observations," *Transportation Research Part B: Methodological*, vol. 173, pp. 313–331, 2023, https://doi.org/10.1016/j.trb.2023.05.004.

[19] B. D. Ziebart, A. L. Maas, J. A. Bagnell, and A. K. Dey, "Maximum entropy inverse reinforcement learning," in *Proceedings of the 23rd AAAI Conference on Artificial Intelligence*, 2008, pp. 1433–1438, https://cdn.aaai.org/AAAI/2008/AAAI08-227.pdf.

[20] J. Neyman and E. S. Pearson, "On the problem of the most efficient tests of statistical hypotheses," *Philosophical Transactions of the Royal Society of London. Series A*, vol. 231, pp. 289–337, 1933, https://doi.org/10.1098/rsta.1933.0009.

[21] P. A. Lopez, M. Behrisch, L. Bieker-Walz, J. Erdmann, Y.-P. Flötteröd, R. Hilbrich, L. Lücken, J. Rummel, P. Wagner, and E. Wiessner, "Microscopic traffic simulation using SUMO," in *21st International Conference on Intelligent Transportation Systems (ITSC)*. IEEE, 2018, pp. 2575–2582, https://doi.org/10.1109/ITSC.2018.8569938.

[22] Q. Ma, X. Guo, W. Zhong, Z. He, Z. Su, W. Ma, and R. Zhong, "City-scale high-resolution traffic datasets with refined networks for hierarchical traffic control," *Scientific Data*, vol. 12, p. 547, 2026, figshare deposit 29925824; https://doi.org/10.1038/s41597-026-06892-2.

[23] R. Cappi, M. Luca, P. Fontolan, N. Navarin, B. Lepri, and A. Sperduti, "A city-scale dataset of traffic flows, travel times, and urban context," *arXiv preprint arXiv:2605.18782*, 2026, https://arxiv.org/abs/2605.18782.

[24] Y. Wang, Y. Chen, G. Li, Y. Lu, Z. He, Z. Yu, and W. Sun, "City-scale holographic traffic flow data based on vehicular trajectory resampling," *Scientific Data*, vol. 10, p. 57, 2023, https://doi.org/10.1038/s41597-022-01850-0.

Appendix

A Code and reproducibility

A.1 Repository structure

The public repository is available at `aaron-bendor/Masters`. The codebase is organised as follows.

`morimura.py` The inverter core: `Inverter` class with softmax-parametric chain, analytic gradients (local + globals), λ-cross-validated fit, and the synthetic reproduction driver.

`congestion_filter.py` Early exploratory experiments: snapshot-level two-regime synthesis and detection.

`per_car_detector.py` Tier-2 per-trajectory log-likelihood-ratio detector.

`sumo_validation/` The SUMO-side machinery: `sumo_to_phase3.py` (network builder, trip reader, feature extractor), `score_seed_big.py` and `contrast_correlation.py` (scorers, with the `-dataset {sumo,xuancheng}` switch), the empirical and backoff 2-step diagnostics (sections 4.3.3 and 5.3.2), and `fd_check_features.py` (the FD gradient gate). Exact script names and log paths are indexed in the SUMO validation README.

`real_data/` The Xuancheng tier: the unmodified `xuancheng.net.xml` from [22], the daily JSON trip files, and `load_xuancheng.py` which provides the loader, the three regime-split helpers, and the OD-rebalancing extension.

`papers/` Source PDFs of all cited works.

`research_logbook.md` The day-by-day project log, with twenty experimental-cycle entries.

A.2 Build and reproduction commands

The project is reproducible from the repository via the README, with the following minimum environment:

```
# Python 3.14, see project README
python3 -m venv .venv
.venv/bin/python -m pip install numpy scipy matplotlib

# Tier 1: synthetic Morimura reproduction (~20 s)
.venv/bin/python morimura.py

# Tier 2: SUMO experiments (requires SUMO_HOME set)
.venv/bin/python sumo_validation/contrast_correlation.py \
    --seed 42 --features real --T 40 --maxiter 1500

# Tier 3: Xuancheng experiments (requires real_data/ populated)
.venv/bin/python sumo_validation/contrast_correlation.py \
    --dataset xuancheng \
    --day 2023-04-17,2023-04-18,2023-04-19,2023-04-20,2023-04-21 \
    --regime_split rush_offpeak --od_match 8 \
    --features real --T 10 --maxiter 1500
```

```
# Stronger-proxy Labour-Day stress test
DAYS=2023-04-28,2023-04-29,2023-04-30,\
2023-04-11,2023-04-12,2023-04-13,2023-04-14,\
2023-04-17,2023-04-18,2023-04-19,2023-04-20,2023-04-21
.venv/bin/python sumo_validation/contrast_correlation.py \
    --dataset xuancheng \
    --day "$DAYS" \
    --regime_split holiday_normal --od_match 64 \
    --features none --T 10 --maxiter 1500
```

B Solver profiling for the $f + g$ extension

This appendix records the implementation and profiling detail behind section 4.3.5. The optional $f + g$ loss switches the full-network fits from the f-only setting ($\gamma = 1$) to the Morimura formulation with hitting rates ($\gamma \in [0, 1)$). Initial SUMO-scale tests suggested infeasibility: three sweep attempts at `maxiter` $\in \{1500, 300, 100\}$ and $|X_o| \in \{601, 120, 72\}$ did not complete in finite wall-clock, with the representative rush-chain fit at `maxiter=1500`, $|X_o| = 601$ running 9.5 hours without an L-BFGS termination signal.

Sparse-LU implementation. `morimura.py` now includes a `hitting_pack_sparse` path that constructs $(I - \beta P_T^{\setminus j})$ in CSC form from the nonzero structure of P_T and factorises it with `scipy.sparse.linalg.splu`. The `grad_log_h` routine takes a generic `solve_fn` callable so the dense and sparse paths share the same gradient code. A finite-difference check on a synthetic $n = 40$ graph with $\gamma = 0.5$ and all four parameter blocks active confirms equivalence: $\max_i |\nabla L^{(i)}_{\text{dense}} - \nabla L^{(i)}_{\text{sparse}}| = 7 \times 10^{-15}$, and analytic-vs-FD relative error is $\leq 10^{-7}$ across sampled dimensions.

SUMO-scale timing. A timed run at seed 42, $|X_o| = 601$, $d = 8{,}392$ gives the intrinsic-chain $f + g$ fit at `maxiter=10` a wall time of 1,787 s, or about 180 s per L-BFGS iteration. This is roughly 8× slower than the dense baseline extrapolated from the 9.5-hour run. The bottleneck is not the sparse LU factorisation itself; it is the solve on the wide dense gradient matrix $V \in \mathbb{R}^{n \times d}$ built inside `grad_log_h`. Dense LAPACK GETRS exploits BLAS3 multi-core kernels on that wide right-hand side, whereas `splu.solve` processes dense right-hand sides much more column-wise. At $d = 8{,}392$, that throughput advantage outweighs the factorisation sparsity gain. The plausible city-scale fix is therefore Sherman–Morrison factor sharing across all $j \in X_o$ (one factor of $(I - \beta P_T)$ per L-BFGS iteration rather than $|X_o|$ related factors), or the natural-gradient route of [1].

Subgraph tractability and γ sensitivity. On the largest strongly-connected component of one Xuancheng K-means zone ($K = 8$, zone 0, post-SCC $n = 194$, $|X_o| = 53$, $d = 586$), sparse-LU is operationally fine: an f-only fit at `maxiter=300` converges in 1.4 s, and $f + g$ fits run in 46–50 s. The binding sensitivity is then the loss weight γ. At $\gamma = 0.5$, floor-dominated empirical g overwhelms L_f and hurts performance; at $\gamma = 0.95$, $f + g$ recovers the full-observation empirical reference within sampling noise (table 11).

Table 11: γ sweep for $f+g$ on the Xuancheng $K=8$ zone-0 subgraph ($n=194$, $|X_o|=53$, $T=10$, `maxiter`=300). The f-only column is the same fit ($\gamma=1$) shown for reference; the empirical 1-step reference on this subgraph is 0.563.

γ	AUC_f	AUC_{f+g}	Δ	note
0.5	0.498	0.488	-0.010	floor-dominated
0.7	0.498	0.487	-0.010	still floor-dominated
0.9	0.498	0.521	$+0.023$	starts to help
0.95	0.498	**0.553**	**+0.055**	matches empirical reference (0.563)
0.99	0.498	0.489	-0.009	essentially f-only

A $\gamma=0.9$ sweep across all eight $K=8$ zones shows that the subgraph lift is not uniform. Restricting to zones with at least 100 test trajectories (zones $0,5,6,7$), the mean Δ_{f+g} is $+0.018$ with standard deviation 0.038: positive in trend, but within noise. The g-floor fraction remains in the 80–90% band across zones, so restricting to subgraphs does not remove the data-density issue. The refined conclusion is therefore: naive sparse-LU does not transfer at city scale, γ tuning is the binding sensitivity where the solver is tractable, and empirical g sparsity persists at every tested scale.

C Tier 1 synthetic reproduction details

The Tier 1 reproduction validates the estimator component of the filter in isolation. A synthetic chain is constructed on $n=100$ states with a random strongly-connected adjacency and mean out-degree ~ 3. The true transition kernel is a softmax of eq. (1); all parameters (ν, ω^{loc}, ω^{glo_1} and ω^{glo_2}) are drawn iid $\mathcal{N}(0,1)$, with $d_T=d_\psi=5$ random global features as in Morimura et al.'s benchmark. The truth is then mixed 70/30 with a Dirichlet(0.3) row distribution to create a mild model-class gap analogous to the real experiments.

For each observation fraction $|X_o|/n \in \{0.05, 0.10, 0.20, 0.35, 0.50, 0.70, 0.90\}$, X_o is sampled randomly and the inverter is fitted on the analytic f and g derived from the true chain at X_o. Tikhonov regularisation is cross-validated over $\{10^{-4}, 10^{-3}, 10^{-2}, 10^{-1}\}$ on a 75/25 split of X_o. The reported metric is

$$\mathrm{RMAE}(\hat{\pi},\pi) = \frac{\frac{1}{|\mathcal{X}\setminus X_o|}\sum_{x\notin X_o}|\hat{\pi}(x)-\pi(x)|}{\frac{1}{n}\sum_{x\in\mathcal{X}}\pi(x)}, \tag{3}$$

the mean absolute error on unobserved states normalised by the average true stationary mass. Each $|X_o|$ value is repeated for 10 trials.

Table 12: Tier 1 reproduction RMAE versus visual estimates from Morimura et al.'s Fig. 2A. Mean over 10 trials, $n = 100$, full reproduction protocol including global features and cross-validated λ.

$\lvert X_o \rvert$	with g	no g	NWKR	paper visual
5	0.97 ± 0.28	1.08 ± 0.45	2.31	~ 0.7
10	0.49 ± 0.13	0.69 ± 0.20	1.59	~ 0.5
20	0.37 ± 0.07	0.75 ± 0.40	1.73	~ 0.4
35	0.26 ± 0.06	0.44 ± 0.07	1.38	~ 0.3
50	0.21 ± 0.07	0.46 ± 0.13	1.67	~ 0.2
70	0.19 ± 0.07	0.41 ± 0.13	1.42	~ 0.15

The reproduction tracks the paper's proposed-method curve within ~ 0.05 RMAE for $|X_o| \in [10, 70]$. The main deviation is $|X_o| = 5$, where the validation split leaves approximately one validation state and therefore over-regularises. The Nadaraya–Watson kernel-regression baseline does not drop as steeply as the paper's comparator, most plausibly because of a comparator-implementation mismatch rather than an inverse-MC implementation error.

Scale-free prediction convention

Morimura et al. write the recovered prediction as $\hat{f}(x) = \hat{c}\hat{\pi}(x)$ with $\hat{c} = \overline{f}_{X_o}$. Taken literally with a probability vector $\hat{\pi}$ summing to one, this under-scales predictions by a factor tied to the number and mass of observed states and collapses the RMAE values towards one. The convention used here is

$$\hat{f}(x) = \frac{\sum_{i \in X_o} f_i}{\sum_{i \in X_o} \hat{\pi}_i} \hat{\pi}(x). \tag{4}$$

The scale calibration is separate from the optimisation objective. The Morimura count loss is a log-ratio loss:

$$L_d(\theta) = \frac{1}{2} \sum_{i,j \in X_o} \left(\log \frac{\pi_i(\theta)}{\pi_j(\theta)} - \log \frac{f_i}{f_j} \right)^2 .$$

Multiplying every observed count by any constant leaves L_d unchanged, so the objective identifies the relative stationary mass on X_o and the chain parameters; the external exposure scale of the count process remains unidentified. Equation (4) is the unique global calibration that makes the fitted stationary surrogate reproduce the observed total mass on X_o:

$$\sum_{i \in X_o} \hat{f}(i) = \sum_{i \in X_o} f_i.$$

It leaves all fitted ratios $\hat{f}(x)/\hat{f}(y) = \hat{\pi}(x)/\hat{\pi}(y)$ unchanged and therefore does not alter the minimiser, gradients, regularisation, or any theoretical claim based on matching stationary ratios. It is also Fisher-consistent for the benchmark's count model: if the data are generated by $f_i = C\pi_i^\star$ for an unknown exposure constant C and the estimator recovers $\hat{\pi} = \pi^\star$, then

$$\frac{\sum_{i\in X_o} f_i}{\sum_{i\in X_o} \hat{\pi}_i}\hat{\pi}(x) = \frac{C\sum_{i\in X_o} \pi_i^\star}{\sum_{i\in X_o} \pi_i^\star}\pi_x^\star = C\pi_x^\star = f_x.$$

The scale-free form therefore preserves the objective's theoretical guarantees while using the only count scale that is identifiable from the observed subset. Applying the literal mean-count convention instead creates a post-estimation scaling error rather than a different inverse-MC solution.

Gradient verification

Across three random parameter draws, including local-only, ϕ_T-only and ψ-only edge cases, the centred finite-difference gradient check returns maximum relative error $\leq 5 \times 10^{-9}$ for every parameter block. This confirms the analytic gradient implementation used in the downstream tiers.

D Tier 2 multistart basin check

This appendix records the multistart audit summarised in section 4.3.2. The multi-seed Tier 2 sweep showed a bimodal gap-closure pattern: random-trip seeds $\{7, 23\}$ landed in a low-gap basin, while seeds $\{42, 101, 2024\}$ landed in a higher-gap basin. The audit asks whether the low-gap result is merely an artefact of the zero-initialisation used by L-BFGS-B.

Both low-basin seeds were rerun with $K = 5$ random restarts per chain (initial $\theta_0 \sim \mathcal{N}(0, I)$, `features=none`, the local-only setting, $T = 20$). The selected restart is the one with the highest held-out validation log-likelihood, compared against the zero-init baseline.

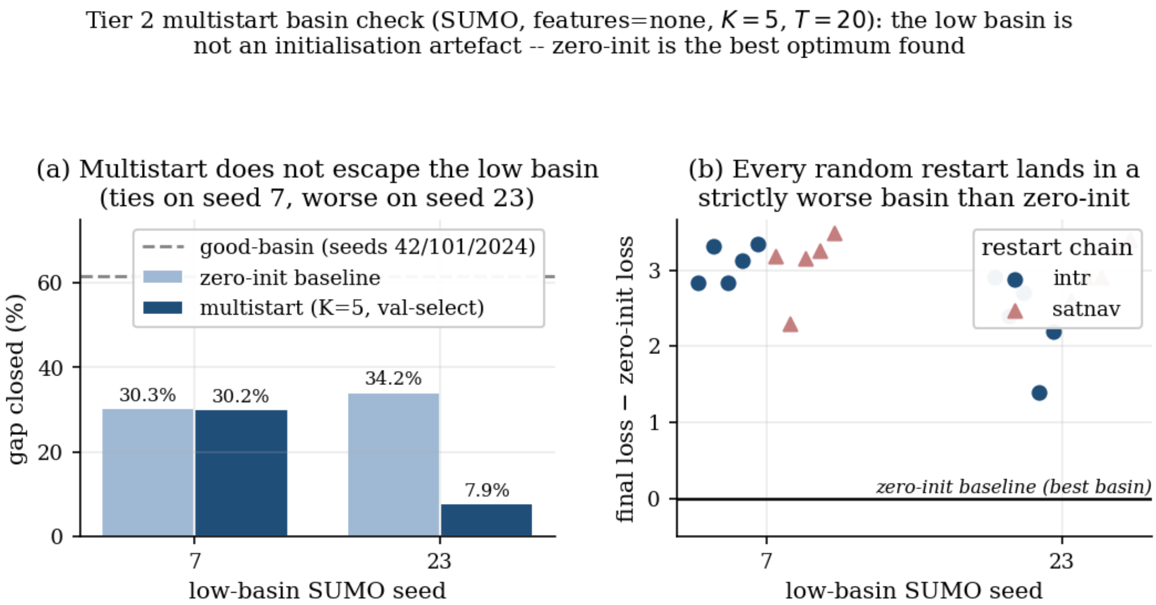

Figure 9: Tier 2 multistart basin check on the two low-basin seeds (`features=none`, $K = 5$, $T = 20$). **(a)** The validation-selected multistart detector ties the zero-init baseline on seed 7 and is markedly worse on seed 23; neither approaches the $\sim 61\%$ good-basin level of seeds $\{42, 101, 2024\}$ (dashed). **(b)** Every one of the ten random restarts (five per regime) converges to a strictly higher final loss than zero-init.

On both seeds, every random restart converged to a worse training loss and a worse held-out validation log-likelihood than the zero-init baseline (fig. 9b). On seed 7, the restarts reach intrinsic-chain loss 44.99–45.5 against the baseline's 42.15; on seed 23, they reach 457–459 against 455.9. The validation-selected multistart detector therefore ties the baseline on seed 7 (30.2% vs. 30.3% gap closed) and is markedly worse on seed 23 (7.9% vs. 34.2%).

The audit does not prove that the low basin is globally optimal. It does show that the standard random-restart recipe does not escape it: zero-init is the best optimum found on these instances, and the cross-seed bimodality should be read as a property of the loss landscapes rather than as a simple starting-point artefact.

E Tier 2 sat-navigation versus congestion decomposition

This appendix records the simulator stress test summarised in section 4.3.4. The motivation is the Tier 3 rush-vs-off-peak proxy: real rush hour may differ from off-peak both because sat-navigation adoption is higher and because congestion changes routing incentives for all drivers. Xuancheng has no counterfactual where one factor is varied while the other is held fixed, but SUMO does.

The two independent levers are the per-vehicle rerouting adoption probability ρ, set by `-device.rerouting.probability`, and the `randomTrips.py` departure period d, set by `-p`. Lower d means more vehicles and therefore more congestion. Five random-trip seeds $\{7, 23, 42, 101, 2024\}$ are swept over three cells per seed: $(d = 0.3, \rho = 0)$ (peak demand, no sat-nav users), $(d = 0.3, \rho = 1)$ (peak demand, full sat-nav), and $(d = 0.8, \rho = 0)$ (off-peak demand, no sat-nav users). Each seed then gives three pairwise AUCs under the same f-only Tier 2 protocol (`features=none`, $T = 20$, $|X_o| = 601$, `maxiter`=1500): `pure_congestion` (demand changes at $\rho = 0$), `satnav_at_peak` (ρ changes at peak demand), and `proxy_diagonal` (the simulated analogue of rush-vs-off-peak).

Table 13: $\rho \times d$ decomposition of the SUMO rush-vs-off-peak proxy signal. Mean $\pm$ sample standard deviation across five SUMO seeds; 95% confidence interval (CI) computed as mean $\pm$ $1.96 \cdot \sigma/\sqrt{5}$.

pair	cells compared	$\mathrm{AUC}_{\mathrm{emp}}$	95% CI	$\mathrm{AUC}_{\mathrm{fit}}$
`pure_congestion`	$(d = 0.3, \rho = 0)$ vs. $(d = 0.8, \rho = 0)$	0.493 ± 0.018	$[0.471, 0.515]$	0.504 ± 0.036
`satnav_at_peak`	$(d = 0.3, \rho = 0)$ vs. $(d = 0.3, \rho = 1)$	$\mathbf{0.780 \pm 0.013}$	$[0.764, 0.796]$	0.647 ± 0.012
`proxy_diagonal`	$(d = 0.8, \rho = 0)$ vs. $(d = 0.3, \rho = 1)$	$\mathbf{0.793 \pm 0.023}$	$[0.764, 0.821]$	0.655 ± 0.026

The `pure_congestion` contrast is statistically indistinguishable from chance: its CI straddles 0.5. The `satnav_at_peak` contrast gives a large robust signal, and the full `proxy_diagonal` is essentially the same magnitude. Decomposing the above-chance empirical AUC of `proxy_diagonal` attributes about 95.6% of the signal to sat-navigation adoption, with the pure-congestion contribution within noise.

This is only a simulator audit. SUMO's $\rho = 0$ vehicles follow precomputed shortest paths and ignore live congestion after departure, whereas real unaided commuters can remember recurring bottlenecks. The result is therefore a lower-bound stress test on the congestion-routing confound. It does not show that the real rush-hour proxy is cleanly separable.

F Metric and method primer

This appendix gives compact reminders for the mathematical objects, optimisation routines and diagnostic metrics used in the main text. The primer fixes the precise meaning of terms that are easy to over-read from their names alone.

F.1 Markov chain and transition kernel

A Markov chain is a stochastic process whose next state depends only on its current state. In this dissertation the states are directed road edges. If a vehicle is currently on edge x, the transition kernel P_T gives the probability of choosing each legal next edge:

$$p_T(x' \mid x) = \Pr[X_{t+1} = x' \mid X_t = x].$$

Each row of P_T is therefore a probability distribution over the legal successors of x: entries are non-negative and sum to one. The "1-step" assumption means that the model ignores the earlier path history once x_t is known. The empirical 2-step diagnostic in section 4.3.3 tests whether this simplification is too severe by allowing the next edge to depend on (x_{t-1}, x_t).

F.2 Stationary distribution, restart probability and hitting rate

The stationary distribution π of a Markov chain is the long-run fraction of time spent in each state if the chain is followed indefinitely. It satisfies

$$\pi^\top P = \pi^\top, \qquad \sum_x \pi(x) = 1.$$

The Morimura framework uses a restarted chain

$$P = \beta P_T + (1-\beta)\mathbf{1}p_I^\top,$$

where β is the probability of continuing along the road-transition kernel and $1-\beta$ is the probability of restarting from p_I. This restart construction makes the chain well behaved and models finite trip lengths.

A hitting rate $g(i,j)$ is a first-passage statistic: starting from observed state i, how likely is the trajectory to hit observed state j later, discounted by the continuation probability β? In principle hitting rates add path-order information beyond simple station counts. In practice they are much noisier on sparse real-world data because there are $|X_o|^2$ station pairs to estimate.

F.3 Inverse Markov-chain problem

The forward Markov-chain problem asks: given P_T, what stationary distribution and hitting rates does it imply? The inverse problem asks the reverse: given partial observations of those statistics at a subset X_o, recover a plausible transition kernel. This is ill-posed without extra structure, because many different kernels can produce similar station counts at a few observation states. Morimura et al.'s method [1] supplies that structure by fitting a softmax-parametric kernel and regularising the parameters.

F.4 Softmax-parametric chain

A softmax turns arbitrary real-valued scores into probabilities. For a road edge x with legal successors $\mathrm{adj}(x)$, the model assigns a score $s(x,x')$ to each possible next edge and sets

$$p_T(x' \mid x) = \frac{\exp(s(x,x'))}{\sum_{y\in \mathrm{adj}(x)} \exp(s(x,y))}.$$

This guarantees a valid probability distribution for every row of P_T. It also lets the model share information through features: the score can combine local edge-specific weights with global features such as road class, speed, lane count or turn angle. In this thesis "features=none" means only local edge weights are used; "features=real" adds shared road-network features as additional parameters.

F.5 Log-likelihood ratio score

Given two fitted regime kernels $\hat{P}_A$ and $\hat{P}_B$, a trajectory $\tau = (x_0, \ldots, x_T)$ is scored by the log-likelihood ratio

$$\Lambda(\tau) = \sum_{t=0}^{T-1} \left[\log \hat{p}_B(x_{t+1} \mid x_t) - \log \hat{p}_A(x_{t+1} \mid x_t)\right].$$

Positive values mean the trajectory is more likely under regime B than regime A according to the fitted chains; negative values mean the opposite. The score is additive, so longer trajectories can accumulate more evidence if the per-step differences are consistent.

F.6 ROC curve and AUC

The receiver-operating-characteristic (ROC) curve evaluates a scalar score across all possible thresholds. For each threshold, it plots the true-positive rate against the false-positive rate. The area under this curve (AUC) is a threshold-free measure of separability:

$$\mathrm{AUC} = \Pr[s(\tau_B) > s(\tau_A)]$$

for a randomly chosen regime-B trajectory and a randomly chosen regime-A trajectory, up to tie handling. An AUC of 0.5 is chance; 1.0 is perfect ranking; values below 0.5 indicate the score is systematically reversed. AUC does not prove correct chain recovery: a classifier can rank regimes well because of aggregate bias, OD mix or optimiser artefacts, which is why the thesis pairs AUC with transition-contrast diagnostics.

F.7 Pearson correlation

Pearson correlation measures linear alignment between two vectors after subtracting their means:

$$r(a, b) = \frac{\sum_k (a_k - \bar{a})(b_k - \bar{b})}{\sqrt{\sum_k (a_k - \bar{a})^2}\sqrt{\sum_k (b_k - \bar{b})^2}}.$$

It lies in $[-1, 1]$: $+1$ means perfect positive linear alignment, 0 means no linear alignment, and -1 means perfect reversal. In this thesis Pearson is computed on stacked legal outgoing-edge contrasts, comparing $\Delta_{\text{fit}} = \hat{P}_B - \hat{P}_A$ with $\Delta_{\text{emp}} = P_{B,\text{emp}} - P_{A,\text{emp}}$. A positive Pearson supports the claim that the fitted model moves probability mass in the same directions as the full-observation empirical chain.

F.8 Cosine similarity and collinearity

Cosine similarity measures directional alignment:

$$\cos(a, b) = \frac{a^\top b}{\|a\| \, \|b\|}.$$

It is insensitive to the vector magnitudes: two vectors pointing in the same direction have cosine $+1$, orthogonal vectors have cosine 0, and opposite directions have cosine -1. Per-row cosine is used at branching states. For a road edge with multiple legal successors, the row vector asks: did the fitted model increase and decrease probability on the same outgoing turns as the empirical contrast? The median row cosine and the fraction of rows with positive cosine are therefore local chain-recovery diagnostics.

F.9 OD matching / OD rebalancing

Origin-destination (OD) matching controls for where trips start and end. Real-world regimes can differ because different people travel to different places before any route-choice difference is considered. The Xuancheng loader clusters network nodes into K spatial zones, bins each trajectory by origin zone and destination zone, and then samples the same number of trajectories from each regime inside every shared OD cell. After this rebalancing, the two regime pools have the same coarse OD distribution, so remaining AUC is closer to a within-OD route-choice contrast. The sweep over K checks whether this conclusion depends on how coarse the OD zones are.

F.10 Bootstrap confidence interval

A bootstrap estimates sampling uncertainty by repeatedly resampling the observed test set with replacement. In the Xuancheng experiments the bootstrap is stratified by regime: each replicate samples the same number of regime-A and regime-B trajectories, recomputes the AUC, and stores the result. The 2.5th and 97.5th percentiles form an empirical 95% confidence interval. These intervals capture test-trajectory sampling noise conditional on the fitted chains, station set, OD match and train/test split. They do not include uncertainty from refitting the inverse-MC model or redrawing X_o.

F.11 L-BFGS-B optimiser, basins and multistart

L-BFGS-B is a quasi-Newton optimiser: it minimises a differentiable loss using gradients and a limited-memory approximation to second-order curvature. The "B" indicates support for simple bound constraints, although the parameter fits here are effectively unconstrained because the softmax maps real-valued scores to valid probabilities.

The inverse-MC loss is non-convex, so L-BFGS-B can converge to different local optima, or basins, depending on the starting point. A multistart check repeats the fit from several random initial parameters and selects the model with the best validation log-likelihood. If multistart changes the AUC or loss substantially, the single-start result may be an optimiser artefact. If multistart fails to improve the detector, as in the Labour-Day Xuancheng stress test, the weak fitted result is less likely to be explained by a poor initialisation alone.

F.12 Ridge regularisation and finite-difference gradient checks

Ridge, or Tikhonov, regularisation adds $\frac{\lambda}{2}\|\theta\|^2$ to the loss. This discourages very large parameter values and helps stabilise an under-constrained inverse problem, especially when only a sparse subset of states is observed.

A finite-difference (FD) gradient check verifies an analytic gradient by perturbing one parameter at a time:

$$\frac{\partial L}{\partial \theta_k} \approx \frac{L(\theta + \epsilon e_k) - L(\theta - \epsilon e_k)}{2\epsilon}.$$

The thesis uses FD checks as safety gates before trusting custom gradient code, especially for the feature-augmented and sparse hitting-rate variants.

F.13 Gap closed

When the full-observation empirical reference has AUC above chance, the gap-closed statistic reports how much of the available empirical signal the fitted detector recovers:

$$\text{gap closed} = \frac{\mathrm{AUC}_{\mathrm{fit}} - 0.5}{\mathrm{AUC}_{\mathrm{emp}} - 0.5}.$$

For example, if the empirical reference is 0.70 and the fitted detector is 0.60, the fitted detector closes $(0.60 - 0.50)/(0.70 - 0.50) = 50\%$ of the above-chance gap. This statistic is undefined or uninformative when the empirical reference is at chance.

F.14 RMAE

Relative mean absolute error (RMAE) is used in the Tier 1 reproduction because Morimura et al.'s original benchmark evaluates stationary-distribution recovery rather than two-regime detection. In this thesis it measures the mean absolute error of the recovered stationary mass on unobserved states relative to the scale of the true held-out counts. Lower is better. It is therefore not directly comparable to AUC: RMAE evaluates reconstruction accuracy, whereas AUC evaluates trajectory ranking between regimes.